9.2 PERMANENT MAGNETS AND CHARACTERISTICS

Permanent-magnet characteristics and their operation within a magnetic circuit are described. The air gap or load line to determine the operating point of the magnet is derived. The energy-density definition and magnet-volume calculation are given to clarify the rudiments of application considerations for the magnet in machine design. The design computation for the rotor magnet in ac machines is outside the scope of this book; the interested reader is directed to the cited references at the end of this chapter.

9.2.1 Permanent Magnets

Materials to retain magnetism were introduced in electrical-machine research in 1950's. There has been a rapid progress in materials in the interim. Materials that retain magnetism are known as *hard magnet* materials. Various materials, such as Alnico-5, ferrites, samarium–cobalt, and neodymium–boron–iron are available as permanent magnets for use in machines. The B–H demagnetization characteristics of these materials are shown in Figure 9.1—for second quadrant only, because the magnets do not have an external excitation once they are magnetized and therefore have to operate with a negative magnetic field strength. That includes the second and third quadrants of the B–H characteristics. It is not usual to encounter the third quadrant in operation; hence, only the second quadrant is considered here. The last three listed materials have straight-line B vs. H characteristics, whereas Alnico-5 has the highest remnant flux density but has a nonlinear B vs. H.

The flux density at zero excitation is known as remnant flux density, B_r, shown in Figure 9.2 for a generic hard permanent magnet. The low-grade magnet has a curve, usually termed a *knee*, at low flux density, and at around this point it sharply dives down to zero flux density and reaches the magnetic field strength of H_{cl} known as coercivity. The magnetic field strength at the knee point is H_k. If the external excitation acting against the magnet is removed, then it recovers magnetism along a line parallel to the original B-H characteristic. In that process, it reaches a new remnant flux density, B_{rr}, considerably lower than the original remnant flux density. The magnet has lost this difference in flux density irretrievably. Even though the recoil line is shown as a straight line, it usually is a loop, and the average of the loop flux densities is represented by the straight line. In the case of a high-grade magnet, the B–H characteristic is a straight line, and its coercivity is indicated by H_{ch}. The line along which the magnet can be demagnetized and restored to magnetism is known as the *recoil line*. The slope of this line is equal to $\mu_0\mu_{rm}$, as derived from the relationship between the flux density and magnetic field intensity, where μ_{rm} is relative recoil permeability. For samarium–cobalt and neodymium–boron magnets, the recoil permeability, μ_{rm}, nearly has a value between 1.03 to 1.1.

9.2.2 Air Gap Line

To find the operating point on the demagnetization characteristic of the magnet, consider the flux path in the machine. The flux crosses from a north pole of the rotor magnet to the stator across an air gap and then closes the flux path from the stator to the rotor south pole via an air gap. In the process, the flux crosses two times the

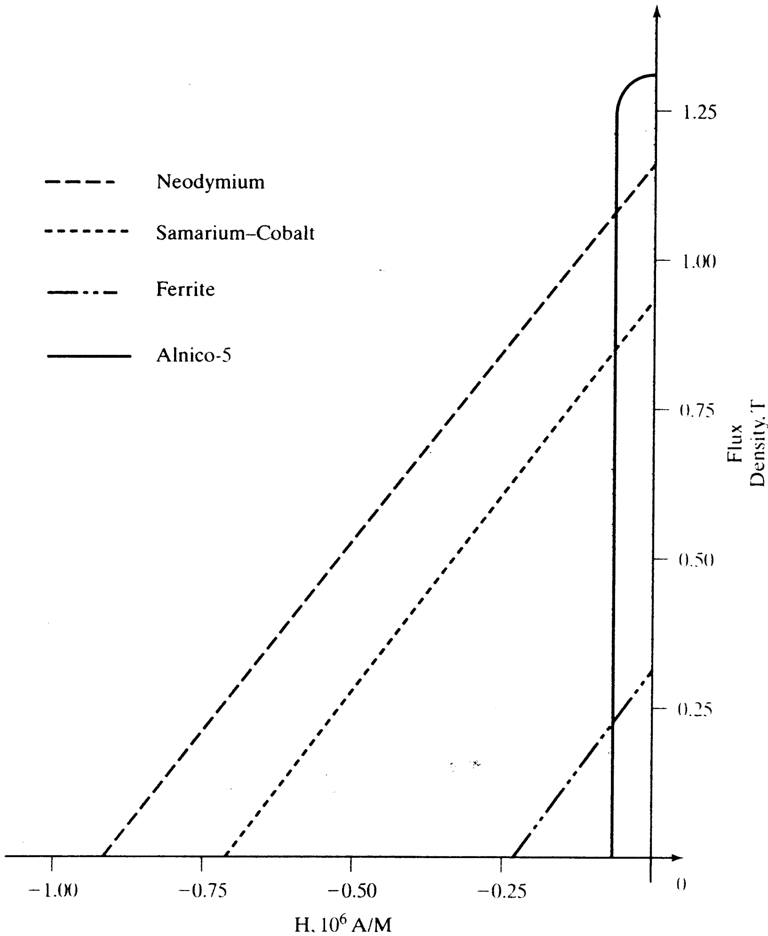


Figure 9.1 Second-quadrant B–H characteristics of permanent magnets

magnet length and two times the air gap, as shown in Figure 9.3. The mmf provided by magnets is equal to the mmf received by the air gap if the mmf requirement of stator and rotor iron is considered negligible. Then,

$$H_m l_m + H_g l_g = 0 \tag{9.1}$$

where H_m and H_g are magnetic field strengths in magnet and air, respectively, and, l_m and l_g are the length of the magnet and air gap, respectively. The operating flux density on the demagnetization characteristic can be modeled, assuming that it is a straight line, as

$$B_m = B_r + \mu_0 \mu_{rm} H_m \tag{9.2}$$

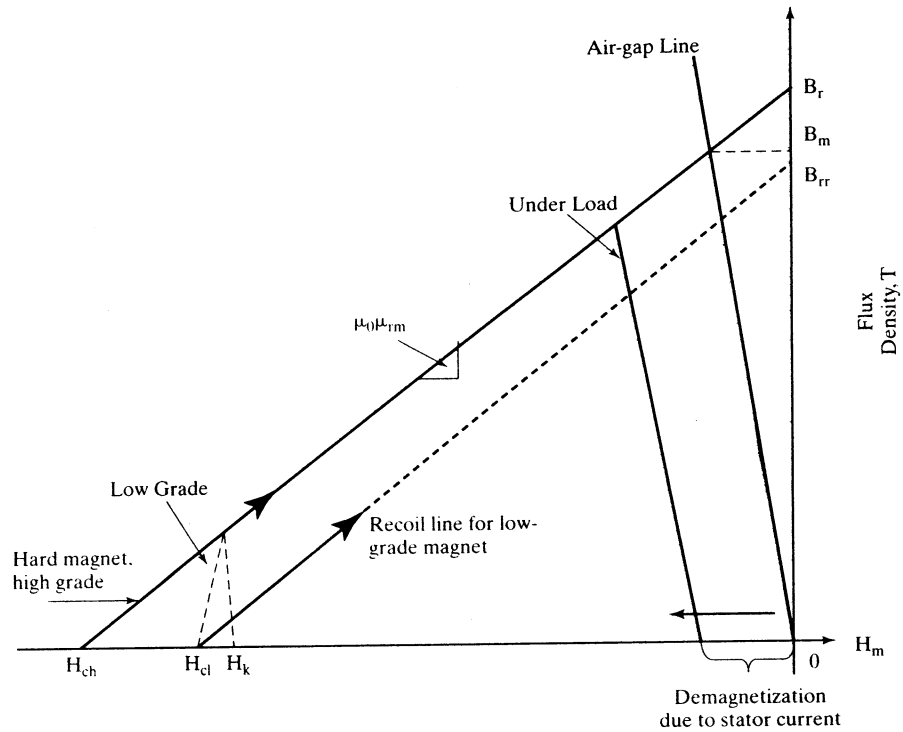


Figure 9.2 Operating point of magnets

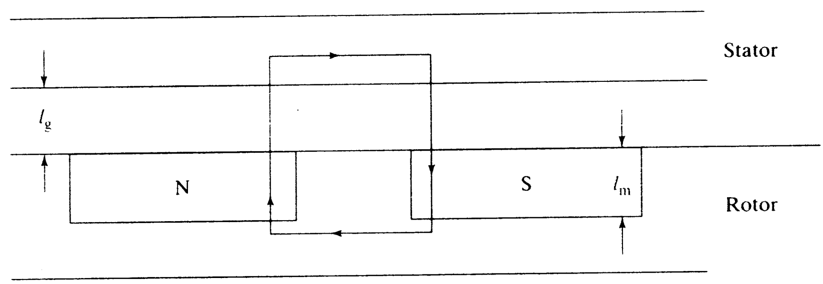


Figure 9.3 Simple layout of the stator and rotor of a machine

Substituting for H_m from (9.1) into (9.2) in terms of H_g, and then writing it in terms of the air gap flux density, which is equal to the magnet flux density, gives the operating magnet flux density:

$$B_m = \frac{B_r}{\left(1 + \frac{\mu_{rm} l_g}{l_m}\right)} \tag{9.3}$$

This clearly indicates that the operating flux density is always less than the remnant flux density, because of the air gap excitation requirement. Note that the excitation requirements of iron and leakage flux are neglected in this conceptual derivation.

The operating point is shown in Figure 9.2, and the line connecting the operating flux density B_m and origin is known as the air gap line or *load line*. The slope of this line is equal to a fictitious permeance coefficient μ_c times the permeance of the air. If the stator is electrically excited, producing demagnetization, then the load line moves toward the left but remains parallel to the original load line, as shown in the figure. The operating flux density is further reduced from B_m. Note that the permeance coefficient is derived for an operating point defined by B_m and H_m as

$$B_m = B_r + \mu_0 \mu_{rm} H_m = -\mu_0 \mu_c H_m \tag{9.4}$$

from which the permeance coefficient is derived as

$$\mu_c = \frac{B_r}{-\mu_0 H_m} - \mu_{rm} = \frac{-\mu_0 \mu_{re} H_m}{-\mu_0 H_m} - \mu_{rm} = \mu_{re} - \mu_{rm} \tag{9.5}$$

where μ_{re} can be considered as external permeability. The variations in the remnant flux density are due to temperature changes as well as to the impact of the applied magnetic field intensity, both of which are induced by external operating conditions, as is clearly seen from this formulation of μ_c, and therefore it is appropriately labeled as external permeability. As the demagnetizing field is introduced by external operating conditions, it is seen that the permeance coefficient will decrease as the external permeability also decreases for that operating point. In hard permanent magnets, the external permeability is on the order of from 1 to 10 in the nominal operating region.

9.2.3 Energy Density

The energy density of the magnet is found as the product of its magnetic field strength and its operating flux density. This measure serves to differentiate magnets for use in machines from those having higher values preferred for high-power-density machines. The peak-energy operating point is optimal from the point of view of magnet utilization. It is found by differentiating the energy density with respect to magnetic field strength and equating to zero to find the magnetic field strength at which it is the maximum. The maximum energy density for a hard high-grade permanent magnet shown in Figure 9.4 is

$$E_{max} = -\frac{B_r^2}{4\mu_0 \mu_{rm}} \tag{9.6}$$

and the flux density at which the maximum energy density available is at $0.5B_r$. The operating line for this flux density is shown as giving the required magnetic field strength. Note that this operating point for maximum energy density requires a considerable amount of demagnetizing field strength from the stator excitation of the machine. Moreover, it will not be possible to maintain this operating point in a variable-speed machine drive; the stator currents will be varying widely over the entire torque–speed region.

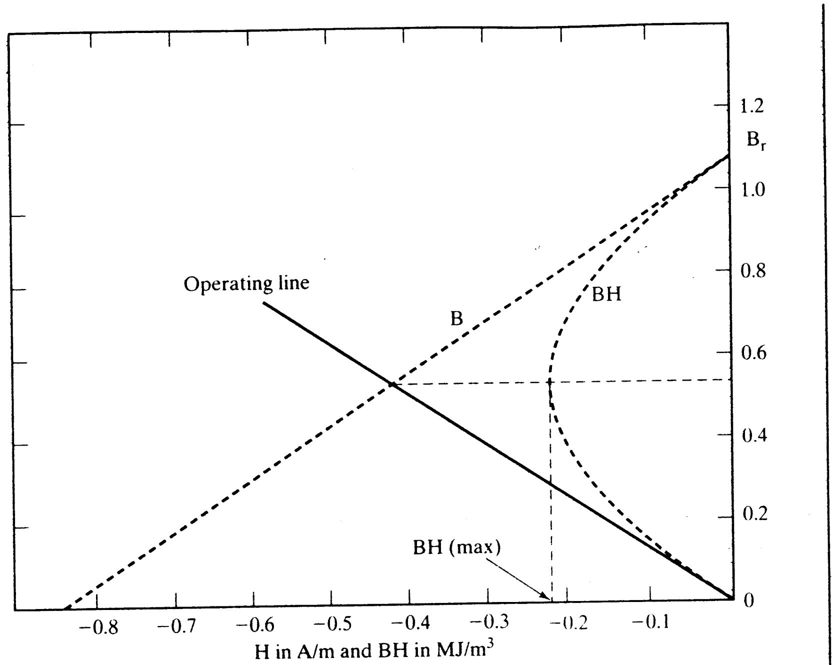


Figure 9.4 Energy product and preferred operating characteristics

9.2.4 Magnet Volume

The magnet volume is found in terms of operating point and air gap volume, as follows:

$$B_g l_g = \mu_0 H_m l_m \tag{9.7}$$

$$B_m A_m = B_g A_g \tag{9.8}$$

From these ideal relationships, magnet volume is

$$V_m = A_m l_m = \left(\frac{B_g A_g}{B_m}\right)\left(\frac{B_g l_g}{\mu_0 |H_m|}\right) = \frac{B_g^2 (A_g l_g)}{\mu_0 |B_m H_m|} = \frac{B_g^2 V_g}{\mu_0 |E_m|} \tag{9.9}$$

where V_g is the air gap volume, E_m is the magnet operating energy density, and A_m and A_g are the magnet and air gap area.

From this relationship, it is inferred that the maximum operating energy density point of the magnet will yield the magnet with the minimum volume and, hence, cost.

9.3 SYNCHRONOUS MACHINES WITH PMs

Prevalent rotor configurations of PM synchronous machines and their impact on the direct and quadrature axis inductances and the distinct differences between the PM brushless dc and synchronous machine are discussed in this section.

9.3.1 Machine Configurations

The permanent magnet (PM) synchronous machines can be broadly classified on the basis of the direction of field flux, as follows:

1. Radial field: the flux direction is along the radius of the machine.
2. Axial field: the flux direction is parallel to the rotor shaft.

The radial-field PM machines are common; the axial-field machines are coming into prominence in a small number of applications because of their higher power density and acceleration. Note that these are very desirable features in high-performance applications.

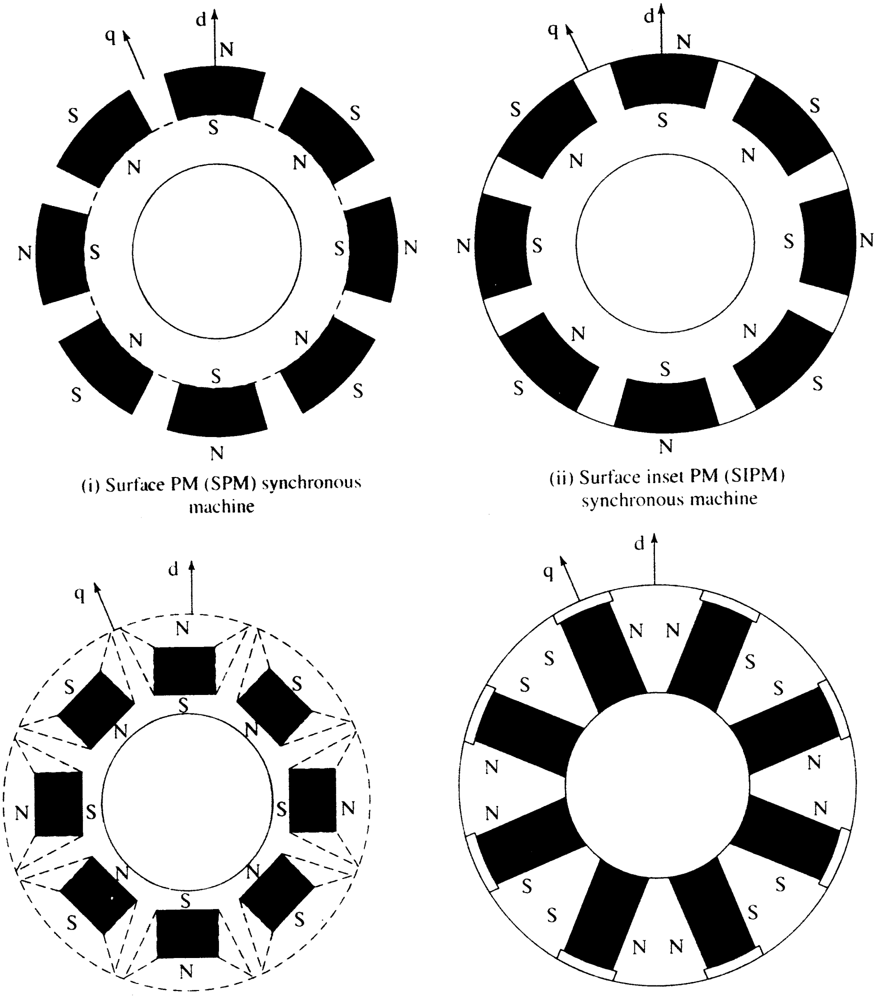
The magnets can be placed in many ways on the rotor. The radial-field versions are shown in Figure 9.5. The high-power-density synchronous machines have surface PMs with radial orientation intended generally for low speed applications, whereas the interior-magnet version is intended for high-speed applications. Regardless of the manner of mounting the PMs, the basic principle of operation is the same. An important consequence of the method of mounting the rotor magnets is the difference in direct and quadrature axes inductance values. It is explained as follows. The rotor magnetic axis is called direct axis and the principal path of the flux is through the magnets. The permeability of high-flux-density permanent magnets is almost that of the air. This results in the magnet thickness becoming an extension of air gap by that amount. The stator inductance when the direct axis or magnets are aligned with the stator winding is known as *direct axis inductance*. By rotating the magnets from the aligned position by 90 degrees, the stator flux sees the interpolar area of the rotor, containing only the iron path, and the inductance measured in this position is referred to as *quadrature axis inductance*. The direct-axis reluctance is greater than the quadrature-axis reluctance, because the effective air gap of the direct axis is multiple times that of the actual air gap seen by the quadrature axis. The consequence of such an unequal reluctance is that

$$L_q > L_d \tag{9.10}$$

where L_d is the inductance along the magnet axis (i.e., direct axis) and L_q is the inductance along an axis in quadrature to the magnet axis.

This is quite contrary to the wound-rotor salient-pole synchronous machine, where the quadrature-axis inductance is always greater than the direct-axis inductance. Note that, in the wound-rotor salient-pole synchronous machine, the direct axis, having the excitation coils, has a small air gap, whereas the quadrature axis has the large air gap.

Figure 9.5(i) shows the magnets mounted on the surface of the outer periphery of rotor laminations. This arrangement provides the highest air gap flux density, but it has the drawback of lower structural integrity and mechanical robustness. Machines with this arrangement of magnets are known as *surface mount PMSMs*. They are not preferred for high-speed applications, generally greater than 3,000 rpm. There is very little (less than 10%) variation between the quadrature- and direct-axis inductances in this machine. This particular fact has consequences for the control, operation, and characteristics of the surface-mount PMSM drives. The details are deferred until later sections.



(i) Surface PM (SPM) synchronous machine

(ii) Surface inset PM (SIPM) synchronous machine

(iii) Interior PM (IPM) synchronous machine

(iv) Interior PM with circumferential orientation synchronous machine

Figure 9.5 (i) Surface-PM (SPM) synchronous machine

Figure 9.5(ii) shows the magnets placed in the grooves of the outer periphery of the rotor laminations, providing a uniform cyclindrical surface of the rotor. In addition, this arrangement is much more robust mechanically as compared to surface-mount machines. The ratio between the quadrature- and direct-axis inductances can be as high as 2 to 2.5 in this machine. This construction is known as inset PM synchronous machine.

Figure 9.5(iii) and (iv) show the placement of magnets in the middle of the rotor laminations in radial and circumferential orientations, respectively. This construction is mechanically robust and therefore suited for high-speed applications.

The manufacturing of this arrangement is more complex than for the surface-mount or inset-magnet rotors. Note that the ratio between the quadrature- and direct-axis inductances can be higher than that of the inset-magnet rotor but generally does not exceed three in value. This type of machine construction is generally referred to as *interior PMSM.*

The inset-magnet construction has the advantages of both the surface- and interior-magnet arrangements: easier construction and mechanical robustness, with a high ratio between the quadrature- and direct-axis inductances, respectively. Many more arrangements of the magnets on the rotor are possible, but they are very rarely used in general industrial practice. Flux-reversal machines with magnets and armature windings on the salient stator poles and salient rotors with no windings or magnets are another possible construction. As they are in the early stages of development, they are not considered any further in this book.

9.3.2 Flux-Density Distribution

The flux plot and flux density vs. rotor position of a surface PM with radial orientation are shown in Figures 9.6 and 9.7, respectively. The dips in the flux density at various points occur because of the slot opening of the stator lamination where the

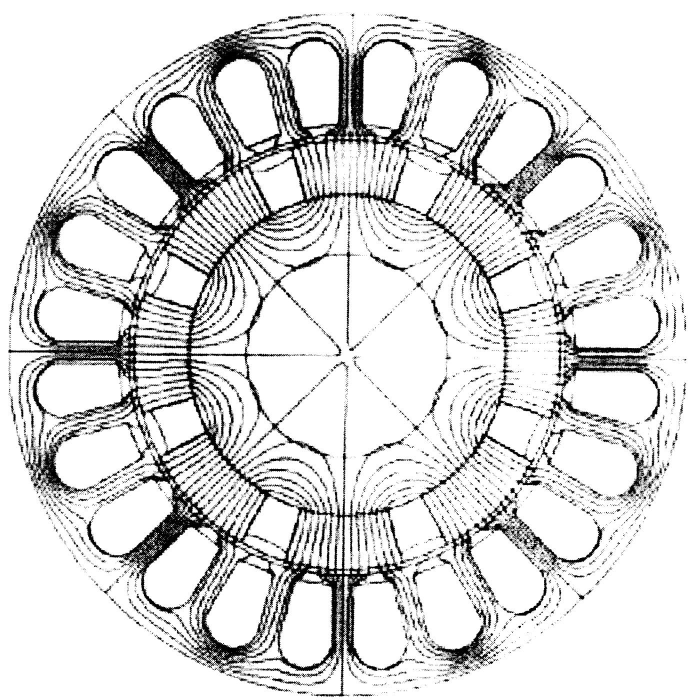


Figure 9.6 Flux distribution in the machine at no-load

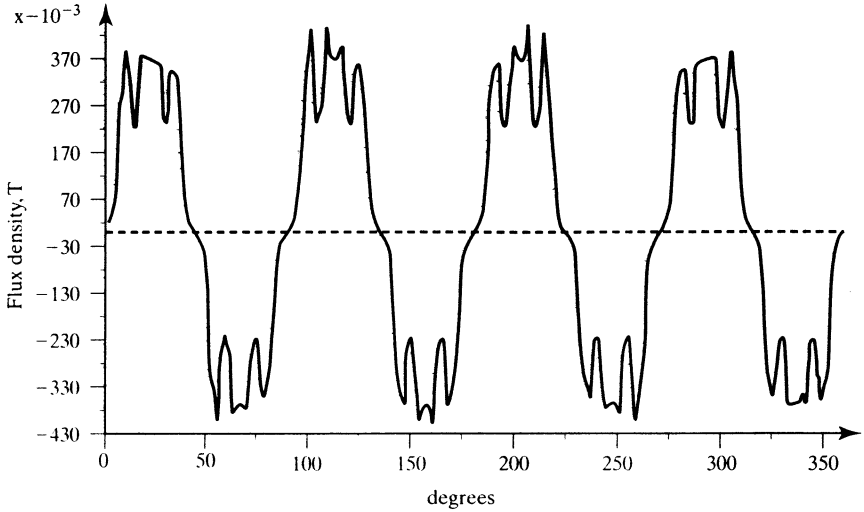


Figure 9.7 Radial air-gap flux-density profile in the machine at no-load

reluctance is much higher and, hence, the flux and its density are lower. The slotting effect also affects the induced emfs in the machine armature, clearly raising design and practical concerns about possible ripple effects. The slotting effect exists regardless of whether the machine is designed with sinusoidal or trapezoidal flux-density distribution. For these distributions, predetermined sinusoidal or rectangular currents are injected to produce the torque. Invariably, that results in ripple air gap torque capable of causing undesirable effects at low speed.

9.3.3 Line-Start PM Synchronous Machines

Some PM synchronous machines are intended and designed for constant-speed applications, to improve efficiency and power factor in comparison to induction and wound-rotor synchronous motors. Such machines have a squirrel-cage winding to provide the torque from standstill to near-synchronous speed. The same cage windings also serve to damp rotor oscillations. Once the motor pulls into synchronism, the cage windings do not contribute to electrical torque, because there are no induced voltages and hence no currents in them at zero slip.

Variable-speed PM synchronous motor drives have no need for the damper windings to offset hunting and oscillation. The damping is provided by properly controlling the input currents from the inverter. This results in a compact and a smaller rotor than that of the machine with damper windings. The way damping is produced in the PM synchronous motor with and without damper windings deserves a comment. The machine with the damper windings operates to suppress the oscillations with no external feedback. The feedback comes internally through the induced emf due to the slip speed in the cage windings. In the inverter-controlled PM drives, the control has to be initiated by an external signal or feedback variable to counter the oscillation. Its dependence on an external feedback loop compromises reliability. Wherever reliable operation regardless of the accuracy in speed control is a major

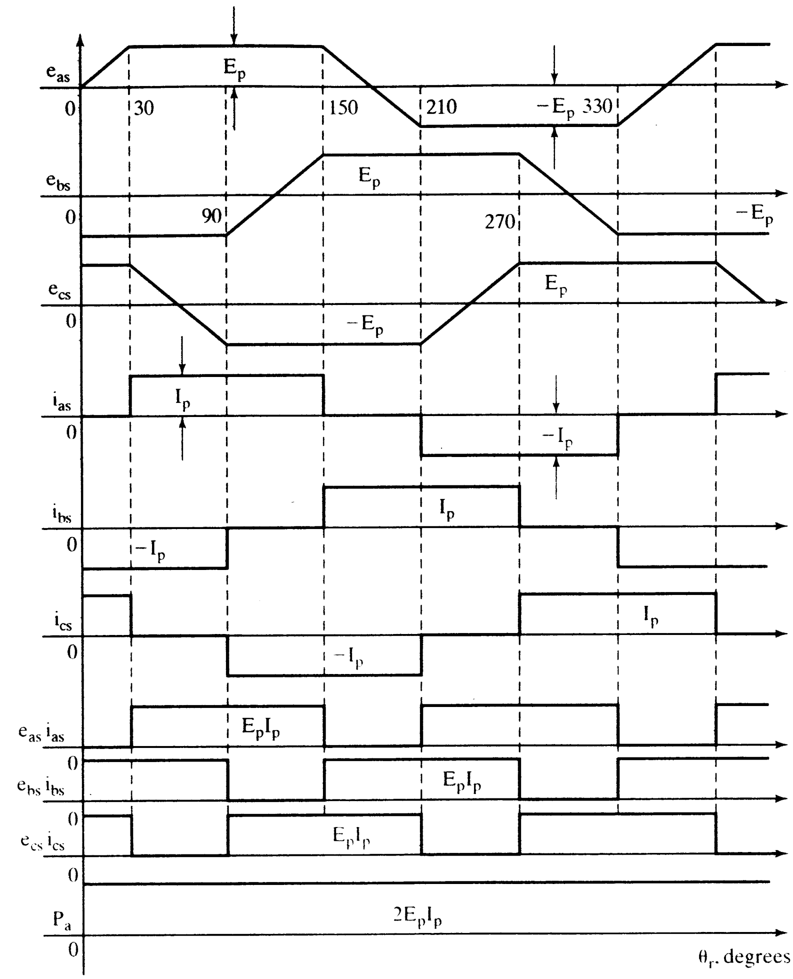


Figure 9.8 PM dc brushless-motor waveforms

concern or requirement, the synchronous motor with damper windings might prove to be an intelligent choice.

9.3.4 Types of PM Synchronous Machines

The PM synchronous motors are classified on the basis of the wave shape of their induced emf, i.e., sinusoidal and trapezoidal. The sinusoidal type is known as PM synchronous motor; the trapezoidal type goes under the name of PM dc brushless machine.

Even though the trapezoidal type of induced emfs have constant magnitude for 120 electrical degrees both in the positive and negative half-cycles, as shown in Figure 9.8, the power output can be uniform by exciting the rotor phases with 120 degrees (electrical) wide currents. The currents cannot rise and fall in the motor windings in zero time; hence, in actual operation, there are power pulsations during the turn-on and turn-off of the currents for each half-cycle. The severity of such pulsations is absent in the sinusoidal-emf-type motors.

PM dc brushless machines have 15% more power density than PM synchronous machines. This can be attributed to the fact that the ratio of the rms value to peak value of the flux density in the PM dc brushless machine is higher than in the sinusoidal PM machine. The ratio of the power outputs of these two machines is derived in the following, based on equal copper losses in their stators.

Let I_{ps} and I_p be the peak values of the stator currents in the synchronous and dc brushless machine. The rms values of these currents are

$$I_{sy} = \frac{I_{ps}}{\sqrt{2}} \tag{9.10}$$

$$I_d = I_p\sqrt{\frac{2}{3}} \tag{9.11}$$

Equating the copper losses and substituting for the currents in terms of their peak currents gives

$$3I_{sy}^2R_a = 3I_d^2R_a \tag{9.12}$$

i.e.,

$$3\left(\frac{I_{ps}}{\sqrt{2}}\right)^2 \cdot R_a = 3\left(\sqrt{\frac{2}{3}}\cdot I_p\right)^2 \cdot R_a \tag{9.13}$$

from which the relationship between the peak currents of the PM synchronous and brushless dc machine is obtained as

$$I_p = \frac{\sqrt{3}}{2}\cdot I_{ps} \tag{9.14}$$

The ratio of their power outputs is obtained from these relationships as follows:

$$\text{Power output ratio} = \frac{\text{PM dc brushless power}}{\text{PM synchronous power}} = \frac{2 \times E_p \times I_p}{3 \times \dfrac{E_p}{\sqrt{2}} \times \dfrac{I_{ps}}{\sqrt{2}}} = \frac{2 \times E_p \times \dfrac{\sqrt{3}}{2} \times I_{ps}}{3 \times \dfrac{E_p \times I_{ps}}{2}} = 1.1547 \tag{9.15}$$

Note that the power output ratio has been derived under the assumption that the power factor of the PM synchronous motor is unity.

From the waveforms of the PM dc brushless motor, it is seen that its control is simple if the absolute position of the rotor is known. Knowing the rotor position amounts to certain knowledge of the rotor field and the induced emf and, hence, instances of applying the appropriate stator currents for control.

9.4 VECTOR CONTROL OF PM SYNCHRONOUS MACHINE (PMSM)

A dynamic model of the PMSM is required to derive the vector-control algorithm to decouple the air gap-flux and torque channels in the drive system. The derivation of the dynamic model is easily made from the dynamic model of the induction machine in flux linkages from Chapter 5.

9.4.1 Model of the PMSM

The two-axes PMSM stator windings can be considered to have equal turns per phase. The rotor flux can be assumed to be concentrated along the *d* axis while there is zero flux along the *q* axis, an assumption similarly made in the derivation of indirect vector-controlled induction motor drives. Further, it is assumed that the machine core losses are negligible. Also, rotor flux is assumed to be constant at a given operating point. Variations in rotor temperature alter the magnet flux, but its variation with time is considered to be negligible. There is no need to include the rotor voltage equations as in the induction motor, since there is no external source connected to the rotor magnets, and variation in the rotor flux with respect to time is negligible.

The stator equations of the induction machine in the rotor reference frames using flux linkages are taken to derive the model of the PMSM. The rotor frame of reference is chosen because the position of the rotor magnets determines, independently of the stator voltages and currents, the instantaneous induced emfs and subsequently the stator currents and torque of the machine. Again, this is not the case in the induction machine: there, the rotor fluxes are not independent variables, they are influenced by the stator voltages and currents, and that is why any frame of reference is suitable for the dynamic modeling of the induction machine. When rotor reference frames are considered, it means the equivalent *q* and *d* axis stator windings are transformed to the reference frames that are revolving at rotor speed. The consequence is that there is zero speed differential between the rotor and stator magnetic fields and the stator *q* and *d* axis windings have a fixed phase relationship with the rotor magnet axis, which is the *d* axis in the modeling.

The stator flux-linkage equations are

$$v^r_{qs} = R_q i^r_{qs} + p\lambda^r_{qs} + \omega_r \lambda^r_{ds} \tag{9.16}$$

$$v^r_{ds} = R_d i^r_{ds} + p\lambda^r_{ds} - \omega_r \lambda^r_{qs} \tag{9.17}$$

where R_q and R_d are the quadrature- and direct-axis winding resistances, which are equal (and hereafter referred to as R_s), and the *q* and *d* axes stator flux linkages in the rotor reference frames are

$$\lambda^r_{qs} = L_s i^r_{qs} + L_m i^r_{qr} \tag{9.18}$$

$$\lambda^r_{ds} = L_s i^r_{ds} + L_m i^r_{dr} \tag{9.19}$$

but the self-inductances of the stator *q* and *d* axes windings are equal to L_s only when the rotor magnets have an arc of electrical 180°. That hardly ever is the case in practice. This has the implication that the reluctances along the magnet axis and the interpolar axis are different. When a stator winding (say *d* axis) is in alignment with the rotor magnet axis, the reluctance of the path is maximum; the magnet reluctance is almost the same as the air gap reluctance, and hence its inductance is the lowest at this time. The inductance then is referred to as the direct-axis inductance, L_d. At this time, the *q* axis winding faces the interpolar path in the rotor, where the flux path encounters no magnet but only the air gaps and iron in the rotor, resulting in lower reluctance and higher inductance. The inductance of the *q* axis winding is L_q at this time. As the rotor magnets and the stator *q* and *d* axis windings are fixed in space, that the winding inductances do not change in rotor reference frames is to be noted. In order, then, to compute the stator flux linkages in the *q* and *d* axes, the currents in the rotor and stator are required. The permanent-magnet excitation can be modeled as a constant current source, i_{fr}. The rotor flux is along the *d* axis, so the *d* axis rotor current is i_{fr}. The *q* axis current in the rotor is zero, because there is no flux along this axis in the rotor, by assumption. Then the flux linkages are written as

$$\lambda_{qs}^r = L_q i_{qs}^r \tag{9.20}$$

$$\lambda_{ds}^r = L_d i_{ds}^r + L_m i_{fr} \tag{9.21}$$

where L_m is the mutual inductance between the stator winding and rotor magnets.

Substituting these flux linkages into the stator voltage equations gives the stator equations:

$$\begin{bmatrix} v_{qs}^r \\ v_{ds}^r \end{bmatrix} = \begin{bmatrix} R_q + L_q p & \omega_r L_d \\ -\omega_r L_q & R_d + L_d p \end{bmatrix} \begin{bmatrix} i_{qs}^r \\ i_{ds}^r \end{bmatrix} + \begin{bmatrix} \omega_r L_m i_{fr} \\ 0 \end{bmatrix} \tag{9.22}$$

The electromagnetic torque is given by

$$T_e = \frac{3}{2}\frac{P}{2}\{\lambda_{ds}^r i_{qs}^r - \lambda_{qs}^r i_{ds}^r\} \tag{9.23}$$

which, upon substitution of the flux linkages in terms of the inductances and currents, yields

$$T_e = \frac{3}{2}\frac{P}{2}\{\lambda_{af} i_{qs}^r + (L_d - L_q) i_{qs}^r i_{ds}^r\} \tag{9.24}$$

where the rotor flux linkages that link the stator are

$$\lambda_{af} = L_m i_{fr} \tag{9.25}$$

The rotor flux linkages are considered constant except for temperature effects. The temperature sensitivity of the magnets reduces the residual flux density and, hence, the flux linkages with increasing temperature. The samarium–cobalt magnets have the least amount of temperature sensitivity: −2 to −3 % per 100°C rise in temperature. Neodymimum magnets have −12 to −13% per 100°C rise in temperature sensitivity; the ceramic magnets have −19% per 100°C rise in temperature sensitivity. Therefore, the temperature sensitivity of the magnets has to be included in the dynamic simulation by appropriately correcting for the rotor flux linkages from their nominal values.

9.4.2 Vector Control

The polyphase PMSM control is rendered equivalent to that of the dc machine by a decoupling control known as vector control. The vector control separates the torque and flux channels in the machine through its stator-excitation inputs. The vector control for PMSM is very similar to that derived in Chapter 8 on vector-controlled induction motor drives. Many variations of vector control similar to that of the induction motor are possible. In this section, the vector control of the PM synchronous machine is derived from its dynamic model. Considering the currents as inputs, the three phase currents are

$$i_{as} = i_s \sin(\omega_r t + \delta) \tag{9.26}$$

$$i_{bs} = i_s \sin\left(\omega_r t + \delta - \frac{2\pi}{3}\right) \tag{9.27}$$

$$i_{cs} = i_s \sin\left(\omega_r t + \delta + \frac{2\pi}{3}\right) \tag{9.28}$$

where ω_r is the electrical rotor speed and δ is the angle between the rotor field and stator current phasor, known as the torque angle.

The rotor field is traveling at a speed of ω_r rad/sec; hence, the q and d axes stator currents in the rotor reference frame for a balanced three-phase operation are given by

$$\begin{bmatrix} i_{qs}^r \\ i_{ds}^r \end{bmatrix} = \frac{2}{3}\begin{bmatrix} \cos\omega_r t & \cos\left(\omega_r t - \frac{2\pi}{3}\right) & \cos\left(\omega_r t + \frac{2\pi}{3}\right) \\ \sin\omega_r t & \sin\left(\omega_r t - \frac{2\pi}{3}\right) & \sin\left(\omega_r t + \frac{2\pi}{3}\right) \end{bmatrix}\begin{bmatrix} i_{as} \\ i_{bs} \\ i_{cs} \end{bmatrix} \tag{9.29}$$

Substituting the equations from (9.26) in (9.28) into (9.29) gives the stator currents in the rotor reference frames:

$$\begin{bmatrix} i_{qs}^r \\ i_{ds}^r \end{bmatrix} = i_s\begin{bmatrix} \sin\delta \\ \cos\delta \end{bmatrix} \tag{9.30}$$

The q and d axes currents are constants in rotor reference frames, since δ is a constant for a given load torque. As these are constants, they are very similar to armature and field currents in the separately-excited dc machine. The q axis current is distinctly equivalent to the armature current of the dc machine; the d axis current is field current, but not in its entirety. It is only a partial field current; the other part is contributed by the equivalent current source representing the permanent magnet field. It is discussed in detail in the section on flux-weakening operation of the PMSM.

Substituting this equation into the electromagnetic torque expression gives the torque:

$$T_e = \frac{3}{2}\cdot\frac{P}{2}\left[\frac{1}{2}(L_d - L_q)i_s^2 \sin 2\delta + \lambda_{af} i_s \sin\delta\right] \tag{9.31}$$

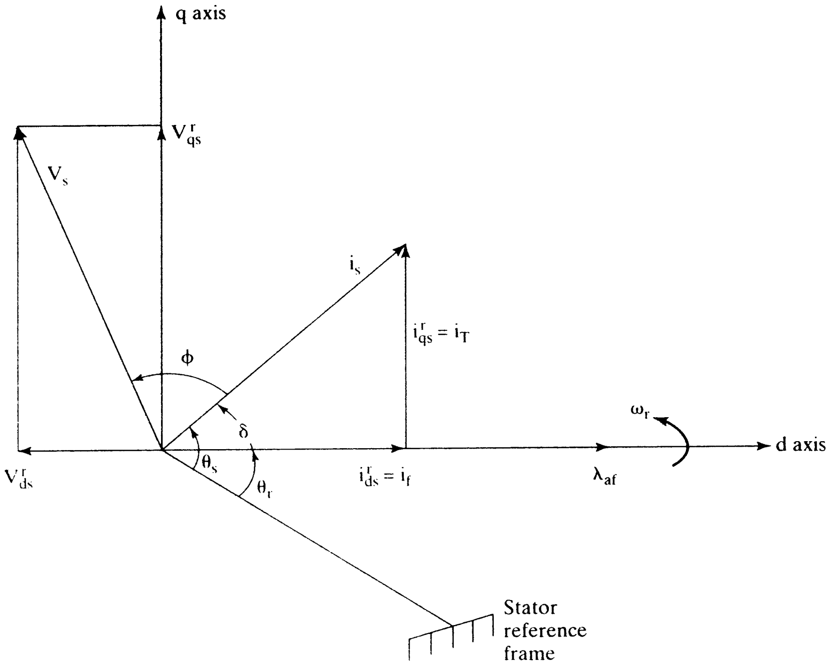


Figure 9.9 Phasor diagram of the PM synchronous machine

For $\delta = \pi/2$,

$$T_e = \frac{3}{2} \cdot \frac{P}{2} \lambda_{af} i_s = K_1 \lambda_{af} i_s, \text{N·m} \tag{9.32}$$

where

$$K_1 = \frac{3}{2} \cdot \frac{P}{2} \tag{9.33}$$

Note that the equation (9.24) is similar to that of the torque generated in the dc motor and vector-controlled induction motor. If the torque angle is maintained at 90° and flux is kept constant, then the torque is controlled by the stator-current magnitude, giving an operation very similar to that of the armature-controlled separately-excited dc motor.

The electromagnetic torque is positive for the motoring action, if δ is positive. Note that the rotor flux linkages λ_{af} are positive. Then the phasor diagram for an arbitrary torque angle δ is shown in Figure 9.9. Note that

$$i_{qs}^r = \text{Torque-producing component of stator current} = i_T \tag{9.34}$$

$$i_{ds}^r = \text{Flux-producing component of stator current} = i_f \tag{9.35}$$

and the torque angle is given by,

$$\theta_T = \delta \tag{9.36}$$

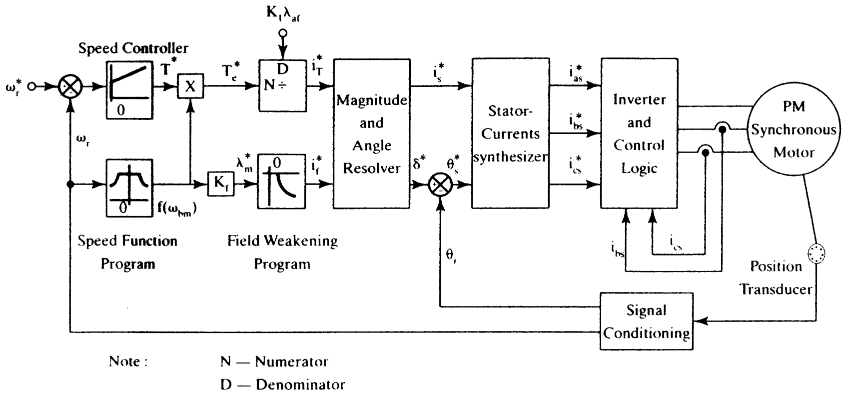


Figure 9.10 Vector-controlled PM synchronous motor drive

The equations (9.34) to (9.36) complete the similarity of the vector-controlled PM synchronous and induction motors. Note that the mutual flux linkage is the resultant of the rotor flux linkages and stator flux linkages. It is then given as

$$\lambda_m = \sqrt{(\lambda_{af} + L_d i^r_{ds})^2 + (L_q i^r_{qs})^2}\ (\text{Wb} - \text{Turn}) \tag{9.37}$$

If δ is greater than $\pi/2$, i^r_{ds} becomes negative. Hence, the resultant mutual flux linkages decrease. This is the key to flux-weakening in the PM synchronous motor drives. If δ is negative with respect to the rotor or mutual flux linkages, the machine becomes a generator.

9.4.3 Drive-System Schematic

From the previous derivations and the understanding gained from them, the vector-controlled PM synchronous motor drive schematic is obtained and is shown in Figure 9.10. The torque reference is a function of the speed error, and the speed controller is usually of PI type. For fast response of the speed, a PID controller is appropriate. The product of torque reference and air gap flux linkage λ^*_m, in p.u., generates the torque-producing component, i^*_T, of the stator current. The reason for this block is to adjust the torque-producing component of the stator current both in the constant-torque and in the constant-power regions of operation. It is proved as follows: The function generator on the speed has the following characteristics:

$$\left.\begin{aligned} f(\omega_{bm}) &= \frac{\omega_b}{\omega_m}; \qquad \pm\omega_b < \omega_m < \pm\,\omega_{max} \\ &= 1; \qquad 0 < \omega_m < \pm\omega_b \end{aligned}\right\} \tag{9.38}$$

and

$$P_a = \omega_m T^*_e = \omega_m f(\omega_{bm}) T^* \tag{9.39}$$

Substituting the function from (9.38) into (9.39) gives the air gap power for the constant-torque region:

$$P_a = \omega_m T^* \tag{9.40}$$

and, for the constant-power operation region, the air gap power is similarly obtained:

$$P_a = \omega_b T^* \tag{9.41}$$

The base speed and T* are constant; hence, the constant-torque and -power modes of operation are implemented with this function-generator block, as seen from the equations (9.40) and (9.41), respectively. Note that subtracting shaft losses from air gap power gives the shaft power output.

The function generator, $f(\omega_{bm})$, sets the reference for the resultant mutual flux linkages, involving a constant K_f. The output of the function generator being in p.u., the value of K_f is unity. The flux-linkages reference, λ_m^*, determines the field current reference, i_f^*, required to counter the flux linkages due to the rotor magnet. Note that, in the field-weakening region, i_f^* is negative; by making this positive, the mutual flux linkages could be strengthened, even though it may lead to saturation and therefore is not usually resorted to in practice. The flux-producing component of the stator-current phasor, i_f^*, can be predetermined and programmed in a function generator, such as in a Read-Only Memory (ROM).

The stator-current phasor magnitude i_s^* and the torque angle δ^* commands are evaluated by

$$i_s^* = \sqrt{(i_f^*)^2 + (i_T^*)^2} \tag{9.42}$$

$$\delta^* = \tan^{-1}\left[\frac{i_T^*}{i_f^*}\right] \tag{9.43}$$

The instantaneous position of the stator-current phasor command is given by

$$\theta_s^* = \theta_r + \delta^* = \omega_r t + \delta^* \tag{9.44}$$

from which the stator-phase current commands are obtained for a balanced three-phase operation by substituting for i_T^* and i_f^* from (9.24):

$$\begin{bmatrix} i_{as}^* \\ i_{bs}^* \\ i_{cs}^* \end{bmatrix} = \begin{bmatrix} \cos\theta_r & \sin\theta_r \\ \cos\left(\theta_r - \frac{2\pi}{3}\right) & \sin\left(\theta_r - \frac{2\pi}{3}\right) \\ \cos\left(\theta_r + \frac{2\pi}{3}\right) & \sin\left(\theta_r + \frac{2\pi}{3}\right) \end{bmatrix} \begin{bmatrix} i_T^* \\ i_f^* \end{bmatrix} = i_s^* \begin{bmatrix} \sin(\theta_r + \delta^*) \\ \sin\left(\theta_r + \delta^* - \frac{2\pi}{3}\right) \\ \sin\left(\theta_r + \delta^* + \frac{2\pi}{3}\right) \end{bmatrix} \tag{9.45}$$

These stator-phase current commands are amplified by the inverter and its logic and fed to the PM synchronous machine. The rotor position is obtained with a position encoder or synchronous resolver. The velocity signal, ω_r, is extracted from the rotor position by using signal processors available commercially at present. In the derivation of stator-current commands, it was assumed that the zero-sequence current is zero, but the derivation is not limited by this fact; it can be included easily by an additional first-order differential equation similar to the stator zero-sequence equation in the induction-machine model.

9.5 CONTROL STRATEGIES

The torque-angle control provides a wide variety of control choices in the PMSM drive system. Some key control strategies are the following:

(i) constant torque-angle control or zero-direct-axis-current control;
(ii) unity power-factor control;
(iii) constant mutual air gap flux-linkages control;
(iv) optimum-torque-per-ampere control;
(v) flux-weakening control.

These control strategies are derived systematically and analyzed in the following, but illustrated for steady-state operation only.

9.5.1 Constant ($\delta = 90°$) Torque-Angle Control

In this control, the torque angle δ is maintained at 90 degrees; hence, the field or direct-axis current is made to be zero, leaving only the torque or quadrature-axis current in place. This is the mode of operation for speeds lower than the base speed. Such a strategy is commonly used in many of the drive systems. The relevant equations of performance in this mode of operation are

$$T_e = \frac{3}{2} \cdot \frac{P}{2}\lambda_{af} \cdot i_{qs}^r = \frac{3}{2} \cdot \frac{P}{2}\lambda_{af} \cdot I_s \tag{9.46}$$

and torque per unit of stator current is constant, given by

$$\frac{T_e}{I_s} = \frac{3}{2} \cdot \frac{P}{2}\lambda_{af} \tag{9.47}$$

and normalized electromagnetic torque is expressed as

$$T_{en} = \frac{T_e}{T_b} = \frac{\frac{3}{2} \cdot \frac{P}{2}\lambda_{af} \cdot I_s}{\frac{3}{2} \cdot \frac{P}{2}\lambda_{af} \cdot I_b} = I_{sn}\,(\text{p.u.}) \tag{9.48}$$

indicating that torque equals the stator current in p.u, which gives the simplest control for implementation in the PMSM drives. Note that I_{sn} is the normalized stator-current phasor magnitude. Relevant equations to determine the steady-state performance of the PMSM drive with this control strategy are derived in the following. The q and d axes voltages, in steady state, are

$$v_{qs}^r = (R_s + L_q p)I_s + \omega_r\lambda_{af} = R_s I_s + \omega_r\lambda_{af} \tag{9.49}$$

$$v_{ds}^r = -\omega_r L_q I_s \tag{9.50}$$

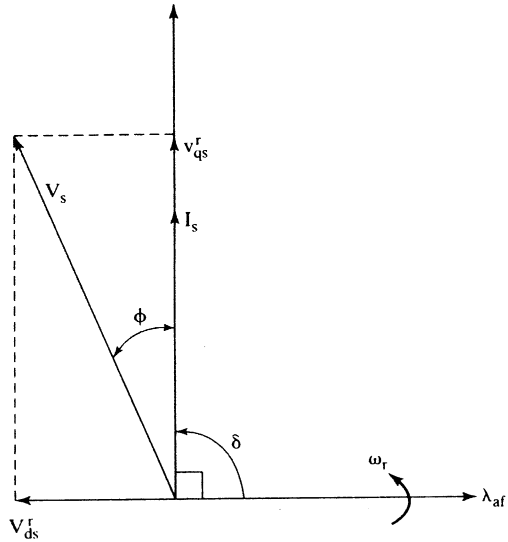


Figure 9.11 Constant-torque-angle control

Note that the rate of change of currents is zero in the rotor reference frames, because the currents are constants in steady state. The magnitude of the voltage phasor is given by

$$V_s = \sqrt{(v_{qs}^r)^2 + (v_{ds}^r)^2} \tag{9.51}$$

and V_s in normalized units is obtained from combining equations (9.49) to (9.51):

$$V_{sn} = \frac{V_s}{V_b} = \frac{V_s}{\omega_b \lambda_{af}} = \sqrt{(\omega_{rn} + R_{sn}I_{sn})^2 + (L_{qn}I_{sn}\omega_{rn})^2} \tag{9.52}$$

From the phasor diagram shown in Figure 9.11 and the axis voltages, power factor is obtained as

$$\cos\phi = \frac{v_{qs}^r}{V_s} = \frac{v_{qs}^r}{\sqrt{(v_{qs}^r)^2 + (v_{ds}^r)^2}} = \frac{1}{\sqrt{1 + \dfrac{(L_{qn}I_{sn})^2}{\left(1 + \dfrac{R_{sn}I_{sn}}{\omega_{rn}}\right)^2}}} \tag{9.53}$$

This equation implies that the power factor deteriorates with increasing rotor speed as well as with increasing stator current. The maximum rotor speed with this control strategy for a given stator current (and neglecting stator resistive drop) is obtained from the voltage magnitude expression (9.52) as follows:

$$\omega_{rn}\,(\max) \cong \frac{V_{sn}(\max)}{\sqrt{1 + L_{qn}^2 I_{sn}^2}} \tag{9.54}$$

where $V_{sn(max)}$ is obtained from the dc-link voltage, V_{dc}, approximately, as

$$V_{sn(max)} = \sqrt{2} \times 0.45 V_{dc} \tag{9.55}$$

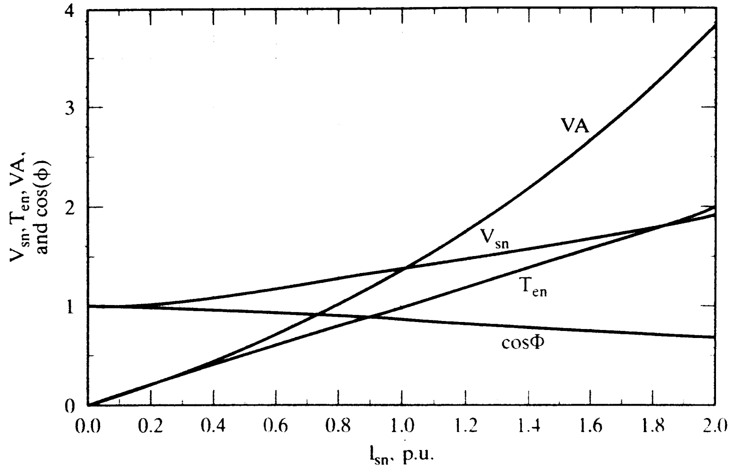


Figure 9.12 Performance characteristics for constant-torque-angle control

assuming that six-step switching is performed in the inverter and neglecting device and cable voltage drops. It is realistic to consider a PWM-based inverter, in which case the available voltage is further reduced by a factor of K_{dr}, usually in the range of from 0.85 to 0.95, giving the voltage phasor as

$$V_{sn(max)} \cong 0.636 K_{dr} V_{dc} \tag{9.56}$$

A PMSM with

$$R_{sn} = 0.1729 \text{ p.u.}, L_b = 0.0129 \text{ H}, L_{qn} = 0.6986 \text{ p.u.}, L_{dn} = 0.4347 \text{ p.u.},$$
$$\omega_b = 628.6 \text{ rad/sec}, V_b = 97.138\text{V}, I_b = 12 \text{ A}$$

is considered for the plotting of performance characteristics utilizing the constant-torque-angle control strategy, as shown in Figure 9.12 for a 1-p.u. rotor speed. Note that the volt-ampere (VA) in p.u. is also plotted as a function of stator current, to assess the VA rating required of the inverter and for the purpose of comparing various control strategies under discussion. The change in power factor from 1 to 0.859 over a zero-to-1 p.u. change in stator current is significant: this creates a need for reactive VA, thus demanding a higher VA from the inverter.

The dc-link voltage required to operate this PMSM at 1-p.u. speed and current is approximately estimated by using equation (9.56). Consider the derating factor to be 0.8, to allow for a margin of voltage for current control. V_{sn} required for this operating point is computed as 1.365 p.u. From this, the required dc-link voltage is obtained as

$$V_{dc} = \frac{1.365}{0.636 \times 0.8} = 2.68 \text{ p.u.} = 2.68 \times 97.138 \text{ V} = 260.6 \text{ V}$$

If current control is not required and six-step voltage-source operation is resorted to in the inverter, the derating factor K_{dr} will go up to from 0.92 to 0.95. This would generate peaky currents, resulting in higher copper losses in the machine, in addition to increased peak inverter-current rating, but the six-step operation will contribute to significant enhancement of the electromagnetic torque and power output, particularly in the flux-weakening region. The maximum speed for this drive (neglecting stator-resistive voltage drop), from equation (9.54), is 1.18 p.u. for 1-p.u. current.

9.5.2 Unity-Power-Factor Control

Unity-power-factor (UPF) control implies that the VA rating of the inverter is fully utilized for real power input to the PMSM. UPF control is enforced by controlling the torque angle as a function of motor variables. The performance equations in this mode of operation are derived and given below.

The q and d axes currents are

$$I_{qs}^r = I_s \sin \delta \tag{9.57}$$

$$I_{ds}^r = I_s \cos \delta \tag{9.58}$$

and normalized torque is obtained as

$$T_{en} = I_{sn}\left\{ I_{sn} \cdot \frac{(L_{dn} - L_{qn})}{2} \cdot \sin 2\delta + \sin \delta \right\} \text{ (p.u.)} \tag{9.59}$$

The q and d axes normalized stator voltages are

$$v_{qsn}^r = \omega_{rn}\left\{ 1 + L_{dn} I_{sn} \cos \delta + \frac{R_{sn} I_{sn}}{\omega_{rn}} \cdot \sin \delta \right\} \text{ (p.u.)} \tag{9.60}$$

$$v_{dsn}^r = \omega_{rn} I_{sn}\left\{ \frac{R_{sn}}{\omega_{rn}} \cdot \cos \delta - L_{qn} \sin \delta \right\} \text{ (p.u.)} \tag{9.61}$$

from which the stator-voltage phasor magnitude is obtained as

$$V_{sn} = \sqrt{(v_{qsn}^r)^2 + (v_{dsn}^r)^2} \text{ (p.u.)} \tag{9.62}$$

The angle between d axis and the resultant voltage V_{sn} is

$$\tan(\delta + \phi) = \frac{v_{qsn}^r}{v_{dsn}^r} \tag{9.63}$$

Since the power factor has to be zero in this control,

$$\phi = 0 \tag{9.64}$$

which gives the following relationship for the torque angle:

$$\tan \delta = \frac{v_{qsn}^r}{v_{dsn}^r} \tag{9.65}$$

Upon substitution of equations (9.60) and (9.61) into (9.65),

$$\frac{\sin \delta}{\cos \delta} = \frac{1 + L_{dn} I_{sn} \cos \delta + \dfrac{R_{sn} I_{sn}}{\omega_{rn}} \cdot \sin \delta}{\dfrac{R_{sn} I_{sn} \cos \delta}{\omega_{rn}} - L_{qn} I_{sn} \sin \delta} \tag{9.66}$$

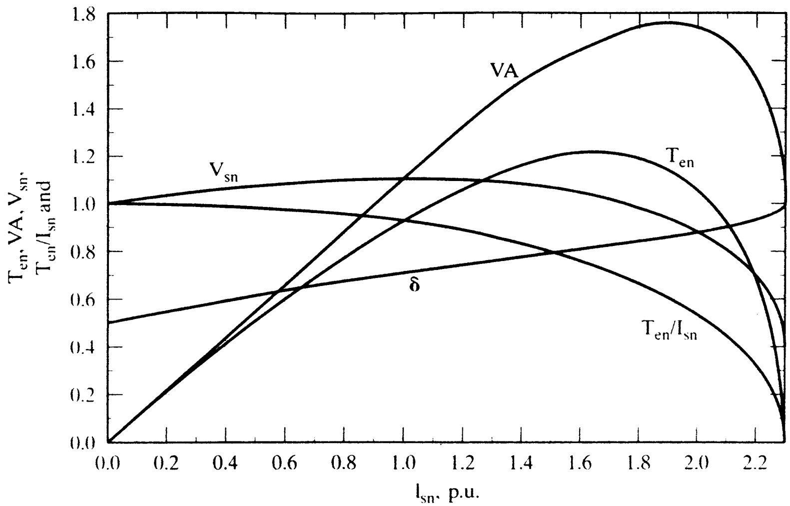


Figure 9.13 Performance characteristics for UPF control

which, upon simplification, gives

$$I_{sn}(L_{qn}\sin^2\delta + L_{dn}\cos^2\delta) = -\cos\delta \tag{9.67}$$

from which torque angle is computed as

$$\delta = \cos^{-1}\left\{\frac{-1 \pm \sqrt{1 - 4L_{qn}I_{sn}^2(L_{dn} - L_{qn})}}{2I_{sn}(L_{dn} - L_{qn})}\right\}\text{(rad)} \tag{9.68}$$

Note that torque angle has to be greater than 90°; hence, depending on the value of the denominator, a positive or negative sign in the above equation has to be employed. Torque angles less than 90° will result in an increase of flux linkages contributing to saturation in the machine; that is undesirable from point of view of losses. This expression provides the enforcement of UPF control; its implementation requires the motor phase current magnitude and motor parameters L_{qn} and L_{dn}. Note that it is independent of rotor speed.

For the same machine cited in the illustration of constant-torque-angle control, the performance characteristics with UPF control are shown in Figure 9.13. The torque per unit current is less than one, indicating that this control strategy is not optimal in terms of torque generation; its efficiency will be reduced by increased copper losses for the generation of same torque. Its overall volt–ampere requirement is only 1.09 p.u., as against 1.355 p.u. for the constant-torque-angle control. The motor-voltage phasor requirement is 1.098 p.u. against 1.365 p.u. for the constant-torque-angle control. This demonstrates that the reserve voltage availability with UPF control would extend the constant torque region, resulting in higher output of the PMSM drive. This feature is very desirable in many applications requiring extended speed range.

9.5.3 Constant-Mutual-Flux-Linkages Control

In this control strategy, the resultant flux linkage of the stator q and d axes and rotor, known as the mutual flux linkage, is maintained constant, most usually at a value equal to the rotor flux linkages, λ_{af}. Its main advantage is that, by the limiting of the mutual flux linkages, the stator-voltage requirement is kept comparably low. In addition, varying the mutual flux linkages provides a simple and straightforward flux-weakening for operation at speeds higher than the base speed. Hence, mutual flux-linkages control is one of the powerful techniques useful in the entire speed range, as against other schemes that are limited to operation lower than the base speed only. The mutual flux linkage is expressed as follows:

$$\lambda_m = \sqrt{(\lambda_{af} + L_d I_{ds}^r)^2 + (L_q I_{qs}^r)^2} \tag{9.69}$$

and, for example, equating it to the rotor flux linkages as

$$\lambda_m = \lambda_{af} \tag{9.70}$$

and substituting for the direct- and quadrature-axis currents in terms of the stator-current phasor and torque angle in the above equations results in

$$I_s = -\frac{2\lambda_{af}}{L_d}\left[\frac{\cos\delta}{\cos^2\delta + \rho^2\sin^2\delta}\right] \tag{9.71}$$

where

$$\rho = \frac{L_q}{L_d} \tag{9.72}$$

Two distinct cases arise here, depending on the saliency ratio ρ. For surface-mounted magnets, ρ is around unity; for buried or interior PM rotor construction it could have values as high as 3. These two cases are analyzed separately in the following.

Case (i): $\rho = 1$

This amounts to a value of torque angle, δ, given by

$$\delta = \cos^{-1}\left\{\frac{-L_d I_s}{2\lambda_{af}}\right\}(\text{rad}) \tag{9.73}$$

Note that the base voltage is defined as

$$V_b = \omega_b \lambda_{af}\,(\text{V}) \tag{9.74}$$

and the base impedance is

$$Z_b = \omega_b L_b = \frac{V_b}{I_b}\,(\Omega) \tag{9.75}$$

Using these for normalization of the motor current given in (9.71) yields the torque angle:

$$\delta = \cos^{-1}\left\{-\frac{L_d I_b \cdot I_s/I_b}{2\lambda_{af}}\right\} = \cos^{-1}\left\{-\frac{I_{sn}L_{dn}}{2}\right\}(\text{rad}) \tag{9.76}$$

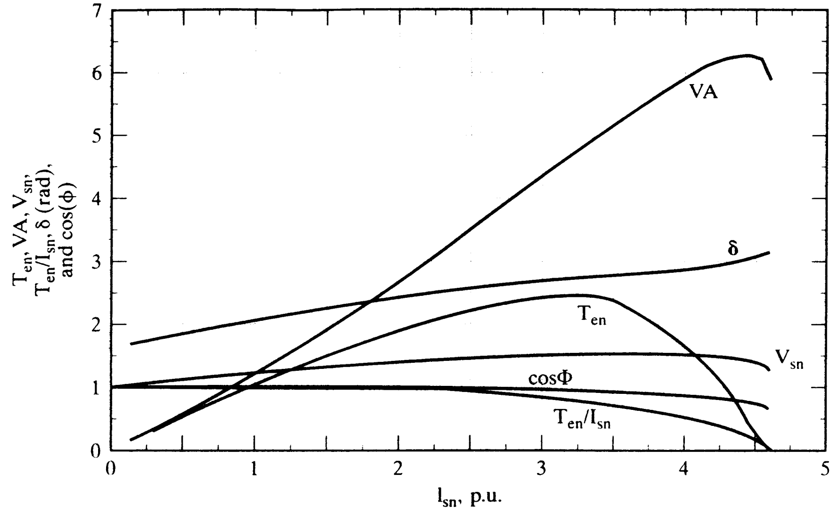


Figure 9.14 Performance characteristics for constant-mutual-flux-linkages control

Case (ii) : $\rho \neq 1$

This yields the torque angle as

$$\delta = \cos^{-1}\left\{\frac{1}{L_{dn}I_{sn}(1-\rho^2)} \pm \sqrt{\left\{\frac{1}{L_{dn}(1-\rho^2)I_{sn}}\right\}^2 - \frac{1}{(1-\rho^2)}}\right\}(\text{rad}) \quad (9.77)$$

The minimum of the two possible values of δ is chosen, so that the demagnetizing current is small. Note also that δ has to be greater than 90°. Otherwise, the *d* axis stator current will strengthen the mutual flux over and above the rotor flux linkages, resulting in saturation of the stator core. The other performance equations are as given in the previous section, and so also are the machine parameters.

The performance characteristics are shown in Figure 9.14. The torque per ampere is slightly less than unity, and stator voltage is approximately 1.17 p.u. (less than is required for constant-torque-angle control), and the VA rating is also less for this control. The power factor is close to unity, up to 1 p.u. of stator current, indicating that this scheme is much closer to UPF control than to constant-torque-angle control.

9.5.4 Optimum-Torque-Per-Ampere Control

A control strategy to maximize electromagnetic torque for unit stator current is valuable from the optimum-machine and inverter-utilization points of view. As in other control strategies, this control strategy is enforced with torque-angle control. The theoretical basis for this control strategy is derived as follows. The electromagnetic torque is

$$T_e = \frac{3}{2}\cdot\frac{P}{2}\left[\lambda_{af}i_s \sin\delta + \frac{1}{2}(L_d - L_q)i_s^2 \sin 2\delta\right] \quad (9.78)$$

which, in terms of normalized units, is

$$T_{en} = i_{sn}\left[\sin\delta + \frac{1}{2}(L_{dn} - L_{qn})i_{sn}\sin 2\delta\right] \text{ (p.u.)} \tag{9.79}$$

where the base torque is defined as

$$T_b = \frac{3}{2}\cdot\frac{P}{2}\lambda_{af}I_b \text{ (N·m)} \tag{9.80}$$

The torque per ampere is given as

$$\left(\frac{T_{en}}{i_{sn}}\right) = \left[\sin\delta + \frac{1}{2}(L_{dn} - L_{qn})i_{sn}\sin 2\delta\right] \tag{9.81}$$

Its maximum is found by differentiating with respect to δ and equating to zero to obtain the condition for δ as

$$\delta = \cos^{-1}\left\{-\frac{1}{4a_1 i_{sn}} - \sqrt{\left(\frac{1}{4a_1 i_{sn}}\right)^2 + \frac{1}{2}}\right\} \text{(rad)} \tag{9.82}$$

where

$$a_1 = (L_{dn} - L_{qn}) = L_{dn}(1 - \rho) \tag{9.83}$$

Only the negative sign in the argument is considered, because δ has to be greater than 90° to reduce the flux in the air gap. To illustrate the superior performance of this control as against constant-torque-angle control, the torque vs. stator current is shown in Figure 9.15. It has 3.2% and 11.05% torque enhancement for 1- and 2-p.u. stator currents, respectively, compared to the constant-torque-angle operation, and this control strategy requires stator voltages of 1.286 and 1.736 p.u., respectively. This is the

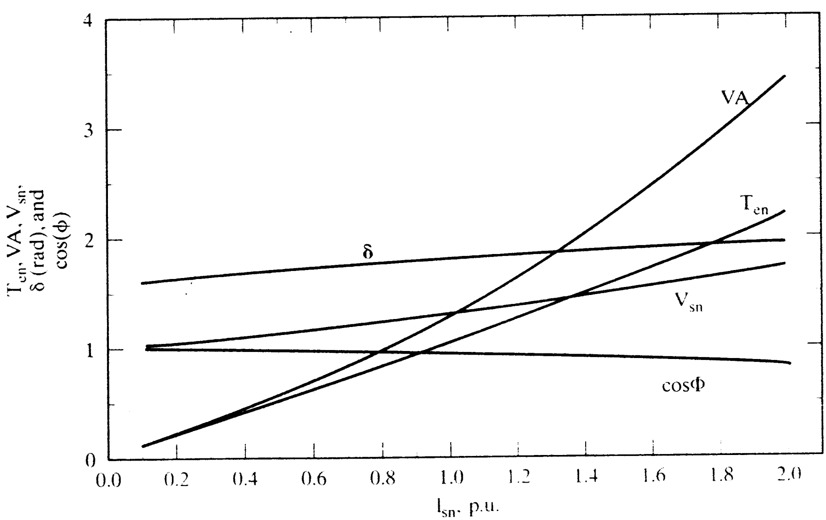


Figure 9.15 Maximum-torque-per-ampere control performance

penalty paid for using this control strategy in terms of the poor inverter utilization, which needs a 3.472-p.u. VA rating at 2-p.u. stator current, as against 2.8 p.u. for constant-mutual-flux-linkages control with a torque derating of 0.065 p.u. only. Note that the torque enhancement could be nearly 100% for ρ =3.5 at a 2-p.u. stator current with this control strategy. This control strategy is, in general, preferred for highly salient machines with $\rho > 2$.

9.6 FLUX-WEAKENING OPERATION

The upper limits placed on the available dc-link voltage and current rating of a given inverter cause the maximum motor-input voltage and current to be limited. The voltage and current limits affect the maximum-speed-with-rated-torque capability and the maximum torque-producing capability of the motor drive system, respectively. It is required and desirable to produce the rated power with the highest attainable speed for many applications, such as electric vehicles, people-carriers in airport lobbies, forklifts, and machine-tool spindle drives. Corresponding to the maximum dc-link voltage and hence to the maximum input machine voltage and rated torque, the machine attains a speed known as the rated speed. Above this speed, the induced emf will exceed the maximum input voltage, making the flow of current into machine phases impractical. To overcome this situation, the induced emf is constrained to be less than the applied voltage by weakening the air gap flux linkages. The flux-weakening is made to be inversely proportional to the stator frequency, so that the induced emf is a constant and will not increase with the increasing speed.

This section considers the operation of the permanent-magnet synchronous motor drives when they are constrained to be within the permissible envelope of the maximum inverter voltage and current to produce the rated power and to provide this at the highest attainable rotor speed. The rated power is intended for steady-state operation; multiple times that is preferred for fast accelerations and decelerations during transient operation. Effective current control during the flux-weakening operation, with high transient capability, is preferable to a saturation of the current loop, resulting in significant harmonic content in the stator currents, resulting in higher torque ripples and higher losses.

Two approaches are considered in this section: (i) The demagnetizing current is predetermined only by rotor speed; this approach is called *direct approach;* (ii) The demagnetizing stator current is derived as a function of not only rotor speed but also the electromagnetic torque. This method is referred to as the *indirect approach*.

9.6.1 Maximum Speed

To understand the scope of the flux-weakening of the PMSM drive, it is essential to find the maximum speed. The maximum speed for a given set of stator voltages and currents is obtained analytically for the purpose of design calculations. The maximum operating speed with zero torque is calculated from the steady-state stator voltage equations as follows. The normalized stator equations in the rotor reference frames are given as

$$\begin{aligned} v_{qsn}^r &= (R_{sn} + L_{qn}p)i_{qsn}^r + \omega_{rn}(L_{dn}i_{dsn}^r + 1) \\ v_{dsn}^r &= -\omega_{rn}L_{qn}i_{qsn}^r + (R_{sn} + L_{dn}p)i_{dsn}^r \end{aligned} \tag{9.84}$$

where the *abc*-to-*qd* transformation valid for voltages, currents, and flux linkages is

$$\begin{bmatrix} v^r_{qsn} \\ v^r_{dsn} \end{bmatrix} = \frac{2}{3}\begin{bmatrix} \cos\theta_r & \cos\left(\theta_r - \frac{2\pi}{3}\right) & \cos\left(\theta_r + \frac{2\pi}{3}\right) \\ \sin\theta_r & \sin\left(\theta_r - \frac{2\pi}{3}\right) & \sin\left(\theta_r + \frac{2\pi}{3}\right) \end{bmatrix}\begin{bmatrix} v_{asn} \\ v_{bsn} \\ v_{csn} \end{bmatrix} \tag{9.85}$$

The steady-state stator-voltage equations are obtained by setting the derivative of the current variables to zero in equation (9.84):

$$\begin{aligned} v^r_{qsn} &= R_{sn}i^r_{qsn} + \omega_{rn}(L_{dn}i^r_{dsn} + 1) \\ v^r_{dsn} &= -\omega_{rn}L_{qn}i^r_{qsn} + R_{sn}i^r_{dsn} \end{aligned} \tag{9.86}$$

when $i^r_{qsn} = 0$, and the stator-voltage phasor is given as

$$v^2_{sn} = {v^r_{dsn}}^2 + {v^r_{qsn}}^2 = \omega^2_{rn}(1 + L_{dn}i^r_{dsn})^2 + R^2_{sn}{i^r_{dsn}}^2 \tag{9.87}$$

from which the maximum speed for a given stator-current magnitude of i^r_{dsn} is

$$\omega_{rn}(\max) = \frac{\sqrt{v^2_{sn} - R^2_{sn}{i^r_{dsn}}^2}}{(1 + L_{dn}i^r_{dsn})} \tag{9.88}$$

Note that the denominator of equation (9.87) has to be positive, giving the condition that the maximum stator current to be applied to counter the magnet flux linkages is

$$i^r_{dsn}(\max) < -\frac{1}{L_{dn}} \tag{9.89}$$

9.6.2 Direct-Flux-Weakening Algorithm

The direct-flux-weakening algorithm finds the demagnetizing component of stator current satisfying the maximum current and voltage limits only. It is very similar to the field control of a separately-excited dc machine, where the field current is determined usually by the speed alone. Such a method has the advantage of simplicity, but it has the disadvantage of not optimizing the stator current by considering the operating torque in the machine. Such an optimization is possible with a torque-request feedforward. The PMSM drive-system control with both constant-torque and constant-power operation is presented in this section. Flux-weakening algorithm, control scheme, controller realization, simulation, and performance are described in this section.

By considering the steady state, the voltage phasor is written (neglecting the resistive terms) as

$$v^2_{sn} = \omega^2_{rn}\{(1 + L_{dn}i^r_{dsn})^2 + (L_{qn}i^r_{qsn})^2\}\ (\text{p.u.}) \tag{9.90}$$

where the voltage phasor, v_{sn}, is defined as

$$v_{sn} = \sqrt{({v^r_{qsn}}^2 + {v^r_{dsn}}^2)}\ (\text{p.u.}) \tag{9.91}$$

The quadrature current i^r_{qsn} can be written in terms of the stator-current phasor and the direct-axis current as

$$i^r_{qsn} = \sqrt{i^2_{sn} - {i^r_{dsn}}^2}\ (\text{p.u.}) \tag{9.92}$$

Substituting equation (9.90) into (9.89) gives the following equation relating the voltage phasor, current phasor, and rotor speed.

$$v_{sn}^2 = \omega_{rn}^2\{L_{qn}^2(i_{sn}^2 - i_{dsn}^{r\,2}) + (1 + L_{dn}i_{dsn}^r)^2\}\ \text{(p.u.)} \tag{9.93}$$

Note that the voltage phasor and current phasor (v_{sn} and i_{sn}, respectively) are the maximum values that could be obtained from the inverter operation. Hence, for the flux-weakening operation, these are considered to be constant. That leads to the appreciation that the equation contains only two variables, ω_{rn} and i_{dsn}^r. Therefore, from one of these two variables the other could be computed analytically. This is the key to the control and operation of the PMSM drive in the flux-weakening region. Further, equation (9.93) is written in terms of i_{dsn}^r and ω_{rn} as

$$v_{sn} = \omega_{rn}\sqrt{(a i_{dsn}^{r\,2} + b i_{dsn}^r + c)}\ \text{(p.u.)} \tag{9.94}$$

where the constants are

$$a = L_{dn}^2 - L_{qn}^2 \tag{9.95}$$

$$b = 2L_{dn} \tag{9.96}$$

$$c = 1 + L_{qn}^2 i_{sn}^2 \tag{9.97}$$

Assuming the rotor speed is available for feedback control and using equations (9.93) to (9.97) gives the d axis stator current, which would automatically satisfy the constraints of maximum stator current, i_{sn}, and stator voltage, v_{sn}. From the stator-current magnitude and the d axis stator current, by using equation (9.92), the maximum permitted q axis stator current could be calculated. The d and q axes currents then determine the stator-phase currents obtained by using the inverse transformation from equation (9.85):

$$\begin{bmatrix} i_{asn} \\ i_{bsn} \\ i_{csn} \end{bmatrix} = \begin{bmatrix} \cos\theta_r & \sin\theta_r \\ \cos\left(\theta_r - \dfrac{2\pi}{3}\right) & \sin\left(\theta_r - \dfrac{2\pi}{3}\right) \\ \cos\left(\theta_r + \dfrac{2\pi}{3}\right) & \sin\left(\theta_r + \dfrac{2\pi}{3}\right) \end{bmatrix} \begin{bmatrix} i_{qsn}^r \\ i_{dsn}^r \end{bmatrix} \tag{9.98}$$

Combining equations $i_{qsn}^r = i_{sn}\sin\delta$ and $i_{dsn}^r = i_{sn}\cos\delta$ with (9.98) yields the normalized stator phase currents via the following relationship:

$$\begin{bmatrix} i_{asn} \\ i_{bsn} \\ i_{csn} \end{bmatrix} = \begin{bmatrix} \sin(\theta_r + \delta) \\ \sin\left(\theta_r + \delta - \dfrac{2\pi}{3}\right) \\ \sin\left(\theta_r + \delta + \dfrac{2\pi}{3}\right) \end{bmatrix} i_{sn} \tag{9.99}$$

The torque angle is obtained as,

$$\delta = \tan^{-1}\left(\frac{i_{qsn}^r}{i_{dsn}^r}\right) \tag{9.100}$$

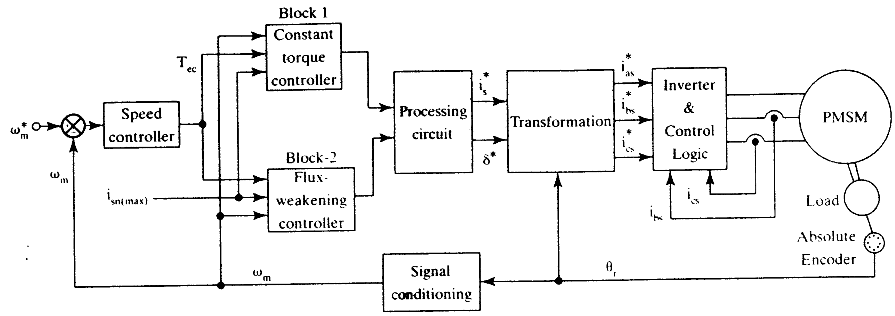


Figure 9.16 Schematic of the PMSM-drive control strategy

Note that the calculated q axis stator current, together with d axis stator current, determines the torque, T_{ef}, that could be produced, and it is then used to modify the torque command, T_{ec}, generated from the speed error in a drive system.

If the torque request, T_{ec}, is more than T_{ef}, the torque calculated by the flux-weakening module, then the final torque request is made equal to the calculated one. If T_{ec} is less than T_{ef}, then the final torque command is maintained at T_{ec}. This final torque request is denoted as T_e^*. From this T_e^*, the required q axis current in the machine could be calculated from the torque equation, since i_{dsn}^r request is known:

$$i_{qsn}^r = \frac{T_e^*}{\{1 + (L_{dn} - L_{qn})i_{dsn}^r\}} \text{ (p.u.)} \tag{9.101}$$

Although the above relationships were derived for steady-state performance, it is to be noted that some voltage reserve is to be available for dynamic control of currents. With a smaller voltage reserve, the current loops will become sluggish; beyond their limit points, they will no longer control the current, thus making the applied voltages six-step. This results in currents rich in harmonics and hence in higher air gap-torque pulsations in the machine.

9.6.2.1 Control scheme. The control scheme for the PMSM drive both in the region of constant torque and in that of flux-weakening could be formulated from the derivations and understanding provided in previous section. Schematically, the control scheme is shown in Figure 9.16. Assuming a speed-controlled drive system, the torque command T_{ec} is generated by the speed error. Depending on the mode of operation, this torque command is processed by Block 1 or Block 2. Block 1 corresponds to the constant-torque-mode controller; Block 2 corresponds to the flux-weakening-mode controller. These blocks are explained in the subsequent paragraphs. The outputs of these controllers are the stator-current-magnitude command and the torque-angle command. They, together with the electrical rotor position, provide the phase-current commands through the transformation block. The current commands are enforced with an inverter by current feedback control, with any one of the current control schemes available. For illustration, pulse-width mod-

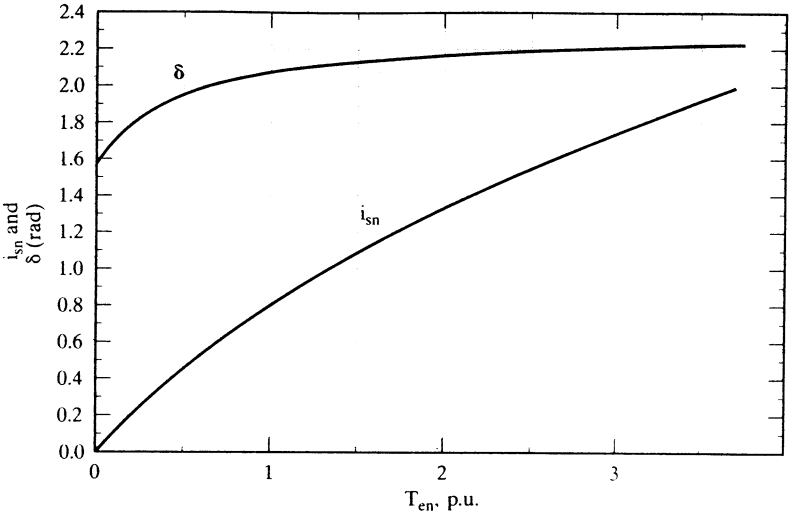


Figure 9.17 Stator-current magnitude and torque angle vs. electromagnetic torque

ulation for the current control is chosen. The rotor position and rotor speed are obtained with an encoder and a signal conditioner, respectively.

9.6.2.2 Constant-torque-mode controller. Block 1 contains the constant-torque-mode controller, with maximum torque per ampere, as explained in 9.4.4. The torque-vs.-stator-current magnitude and torque angle are computed from equations (9.81) and (9.82). Note that, for easier computation, the stator-current magnitude is varied and the torque angle and torque are evaluated. For implementation, note that the inverse is needed, i.e., given the torque request, the stator-current magnitude and torque angle are to be made available. That could be accomplished by curve-fitting the characteristics and programming it in memory for retrieval and use. The characteristics of stator-current magnitude and torque angle vs. torque are shown in Figure 9.17.

These characteristics, for the drive system under illustration, are curve-fitted by the following expressions with minimum error:

$$i_{sn} = 0.01 + 0.954T_{en} - 0.189T_{en}^2 + 0.02T_{en}^3 \tag{9.102}$$

$$\delta = 1.62 + 0.715T_{en} - 0.3T_{en}^2 + 0.04T_{en}^3 \tag{9.103}$$

Algorithm and procedure for obtaining a curve fit can be obtained from a standard textbook on numerical analysis. The detailed schematic of the torque-mode controller, then, takes the form shown in Figure 9.18. The equations (9.102) and (9.103) can be realized in the form of tables to realize Block 1. The speed signal determines the mode of operation of the drive system. In the torque-control mode of operation, if the rotor speed is less than the base speed, then it enables Block 1 in the form of onward transmission of the torque-command signal, T_{ec}. The torque signal is limited

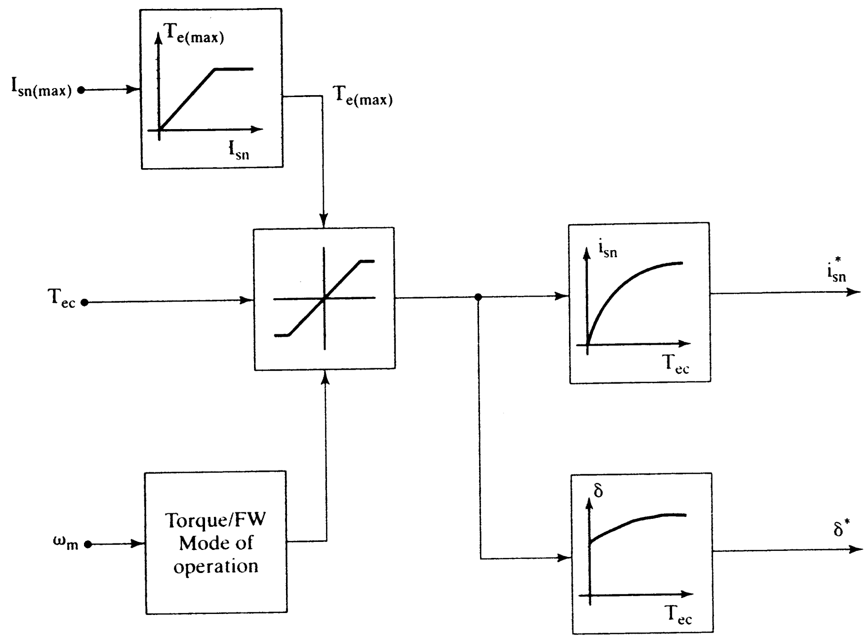


Figure 9. 18 Schematic of constant-torque-mode controller

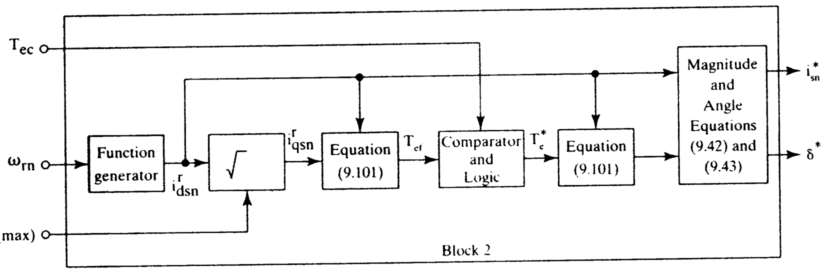


Figure 9.19 Schematic realization of the flux-weakening controller

by the maximum torque that could be generated with the maximum permissible stator-current phasor. Its magnitude can be a variable, depending on whether the drive is in steady state or in intermittent peak operation. Then the resulting torque signal provides the stator-current-magnitude and torque-angle commands from the memory that may have characteristics given in equations (9.102) and (9.103). For regenerative action, the torque angle is negative.

9.6.2.3 Flux-weakening controller. From the flux-weakening algorithm, the flux-weakening controller emerges, which could be schematically captured as shown in Figure 9.19. The inputs to this module are three variables: the torque request, rotor speed, and maximum permissible stator current. The module's outputs are the stator-current-magnitude request and the torque-angle request.

The rotor speed determines the d axis stator current request through the equation (9.94) in a slightly modified form:

$$i^r_{dsn} = \frac{\left(-b + \sqrt{\left\{b^2 - 4a\left(c - \frac{v^2_{sn}}{\omega^2_{rn}}\right)\right\}}\right)}{2a} \tag{9.104}$$

where *a*, *b*, and *c* are given in equations (9.95) to (9.97).

Note that constant *c* is dependent on the maximum stator-current limit. This could either be programmed or be captured in a tabular form for retrieval and use in the feedback-control computations. This d axis current request, indicated with an asterisk in Figure 9.19, then, along with the maximum stator current, determines the permissible quadrature-axis current, i^r_{qsn}.

This *q* axis current, with the d axis current, determines the maximum electromagnetic torque allowed, T_{ef}, within maximum voltage and current constraints. This is compared with the torque request T_{ec}, generated by the speed error. Then the following logic is applied to find the torque-command input to compute the command *q* axis current:

$$\begin{aligned} &\text{If} \quad T_{ec} > T_{ef}, \text{ then } T^*_e = T_{ef} \\ &\text{If} \quad T_{ec} < T_{ef}, \text{ then } T^*_e = T_{ec} \end{aligned} \tag{9.105}$$

This logic submodule adjusts the torque request depending on the load and maximum capability of the motor drive system as a function of the rotor speed. From this final torque request, T^*_e, the stator q axis current is computed by using equation (9.93). The direct- and quadrature-axis stator-current requests are then used to calculate the stator-current phasor magnitude and torque-angle requests. The torque-mode or flux-weakening controller module is chosen, based on the rotor speed being lower or higher than its base value, respectively.

9.6.2.4 System performance. The drive-system performance with the strategy incorporating both the optimized constant-torque mode and flux-weakening controllers is modeled and simulated to evaluate its performance. The PWM frequency of the inverter is set at 5 kHz. The dc-link voltage is 280 V, and the load torque is zero. To prove the operation in the four-quadrant operation, a step speed command from 7 p.u. to −7 p.u. is given, and various machine and control variables are viewed. The machine variables of interest are the torque and speed. Likewise, the control variables, such as speed and torque request are monitored. The simulation results are shown in Figure 9.20. The torque command follows the maximum trajectory as a function of speed, and the actual torque very closely follows its command. The power trajectory is maintained at its set maximum in the flux-weakening mode, and the drive speed envelope is smooth during the entire speed of operation. Note that the base speed is 1 p.u.; beyond it, flux-weakening is exercised in the drive system.

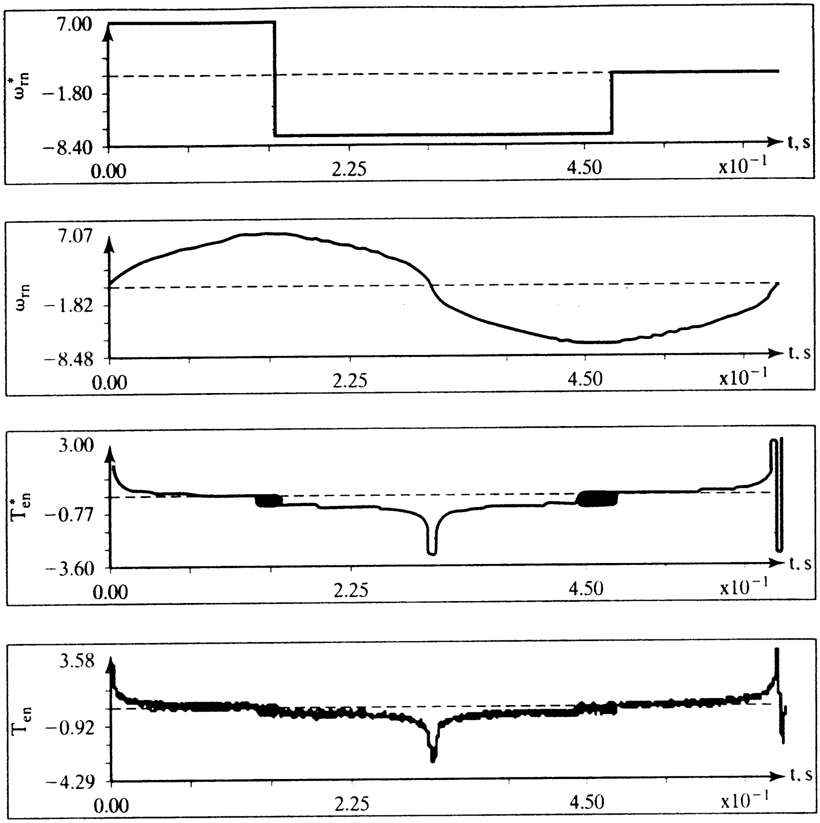


Figure 9.20 Simulation results

Example 9.1

A PMSM has the following parameters:

$$R_{sn} = 0.173 \text{ p.u}, L_{dn} = 0.435 \text{ p.u}, L_{qn} = 0.699 \text{ p.u}, V_{sn} = 1.45 \text{ p.u}, I_{sn} = 1 \text{ p.u.}$$

Find (i) the maximum speed with and without neglecting stator resistances and (ii) the steady-state characteristics in the flux-weakening region.

Solution

Part (i): Maximum Speed

$$\text{Maximum speed} = \frac{\sqrt{V_{sn}^2 - (R_{sn}I_{sn})^2}}{(1 + L_{dn}I^r_{dsn})} \text{(p.u.)}$$

Note that $I^r_{dsn} = -I_{sn}$; hence,

$$\omega_{rn}(\max) = \frac{\sqrt{1.45^2 - 0.173^2}}{1 - 0.435} = 2.547 \text{ p.u.}$$

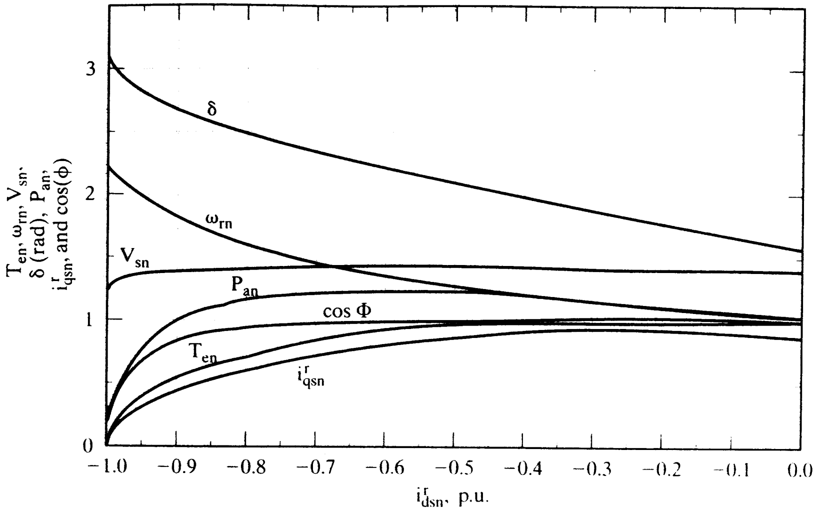


Figure 9.21 Performance characteristics of the PMSM in the flux-weakening region

Neglecting stator resistance gives the maximum speed of the PMSM:

$$\omega_{rn}(\max) = \frac{V_{sn}}{(1 - L_{dn}I_{sn})} = \frac{1.45}{1 - 0.435} = 2.565 \text{ (p.u.)}$$

The effect of the stator resistance is usually negligible in the maximum-speed prediction, but not for small machines, because they have proportionally larger stator resistances compared to large machines.

Part (ii): Steady-State Performance Computation in the Flux-Weakening Operation

The steady-state flux-weakening performance is computed from the following steps.

Step 1: For computing I^r_{dsn} during field-weakening when $I^r_{qsn} = 0$, the effect of resistive drops is, in general, as much as $I_{sn}R_{sn}$ (approximately): hence, reduce V_{sn} to approximately 1.25 p.u.

Step 2: With $V_{sn} = 1.25$ p.u., compute I^r_{dsn} from

$$V_{sn} = \omega_{rn}\sqrt{aI^{r\,2}_{dsn} + bI^r_{dsn} + c}$$

for each value of increasing ω_{rn}. Select the smaller of the roots for I^r_{dsn}. From this, compute I^r_{qsn} with I_{sn} set at 1 p.u.

Step 3: Compute V^r_{qsn}, V^r_{dsn}, and V_{sn} to check on whether it is equal to or less than the original assigned value of 1.45 p.u. Otherwise, go to step 2 to give a smaller value of V_{sn}.

Step 4: Compute voltage-phasor angle and current-phasor angle, hence power-factor angle, torque, and power output, all in p.u.

The performance characteristics are plotted and shown in Figure 9.21.

9.6.3 Indirect Flux-Weakening

An alternative flux-weakening control strategy is based on controlling directly the mutual flux linkage, which is inversely proportional to electrical rotor speed. This is very similar to controlling the field flux of a separately-excited dc machine. The mutual flux linkage is defined as

$$\lambda_{mn} = \sqrt{(L_{dn}i_{dsn} + 1)^2 + (L_{qn}i_{qsn})^2} \tag{9.106}$$

with the assumption that the base flux linkage is equal to the rotor flux linkages. Neglecting the stator resistance voltage drops gives the stator-voltage phasor in terms of the mutual flux linkages and electrical rotor speed by using the equation (9.90) as

$$v_{sn} = \omega_{rn}\lambda_{mn} \tag{9.107}$$

This implies that, during flux-weakening, when the dc link voltage and hence the stator-voltage-phasor magnitude is constant, the mutual flux linkage has to be decreased inversely proportional to the rotor electrical speed. Such control is straightforward and simple in the high-speed operational region, when compared to other schemes in the literature. To implement this, the independent references to be commanded are the mutual flux linkages and electromagnetic torque. In order to incorporate any control strategy (such as the maximum torque per ampere, constant air gap flux linkages, unity power factor, or constant torque angle) below the base speed region, the corresponding mutual flux linkages are calculated and preprogrammed in the mutual-flux-linkages controller. For the region above the base speed, the mutual flux linkage is made to be inversely proportional to the rotor electrical speed in order to work within the limited dc-link voltage. The machine will be operated within its maximum torque capability for a given mutual-flux-linkages command.

9.6.3.1 Maximum permissible torque. The maximum electromagnetic torque for a given mutual flux linkage is found as follows. The torque in terms of motor parameters, mutual flux linkages, and d axis stator current is given by

$$T_{en} = \frac{(L_{dn} - L_{qn})i_{dsn}^r + 1}{L_{qn}}\sqrt{\lambda_{mn}^2 - (1 + L_{dn}i_{dsn}^r)^2} \tag{9.108}$$

By differentiating this with respect to the d axis stator current and equating to zero, the condition for maximum permissible electromagnetic torque for a given mutual flux linkage is found. Substituting this condition for the d axis stator current in (9.108) yields the maximum permissible electromagnetic torque for the given mutual flux linkages.

Figure 9.22 shows such a relationship for $L_{qn} = 0.699$ and $L_{dn} = 0.434$. It can be seen that, for the given system parameters, the T_{enm}^* vs. λ_{mn}^* relationship is nearly linear. This means that a simple first- or second-order polynomial can be used to implement this relationship. In general, the function T_{enm}^*-vs.-λ_{mn}^* can be computed off-line for the appropriate range of λ_{mn}^* and subsequently incorporated into the system as a simple lookup table.

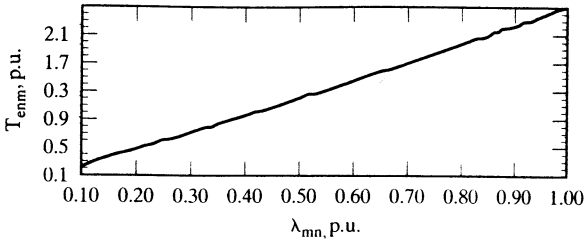


Figure 9.22 Maximum permissible torque command as a function of flux command

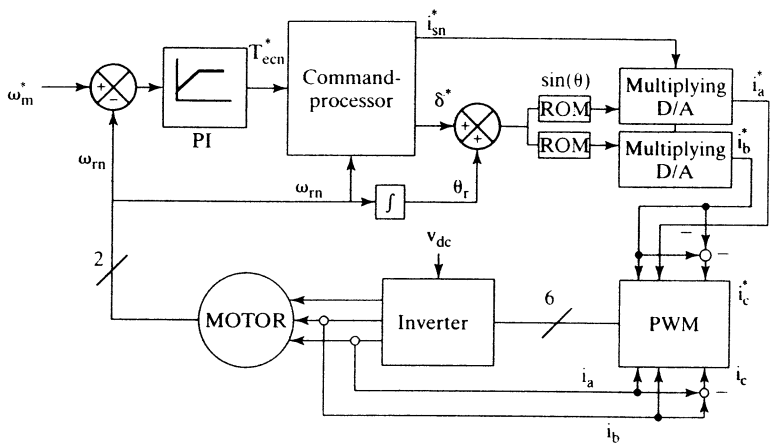


Figure 9.23 The speed-control system with mutual-flux-linkages-based controller

9.6.3.2 Speed-control scheme. The mutual-flux-linkage-weakening and torque controls are incorporated in a speed-control system as shown in Figure 9.23. T_{ecn}^* is the output of the speed PI controller. The command-processor weakens T_{ecn}^* in inverse proportion to speed when the speed is higher than 1 p.u., in order to limit the machine's power. Note that the torque command is limited to its maximum permissible value, which depends on the mutual-flux-linkage command. The command-processor also generates the mutual-flux-linkage command. The mutual-flux-weakening operation begins at base speed, which could be less than or equal to one p.u.. The outcomes, T_{en}^*, along with the commanded mutual flux linkage, λ_{mn}^*, are provided to the current and angle resolver to generate the appropriate commands for the stator current, i_{sn}^*, and its angle, δ^*. The angle δ^* is added to the rotor's absolute angle to provide the desired current angle with respect to the stator's reference frame. The $\{i_{sn}^*, \delta^*\}$ pair is itself a command input to a PWM current controller. Figure 9.24 shows the operations performed by the command processor.

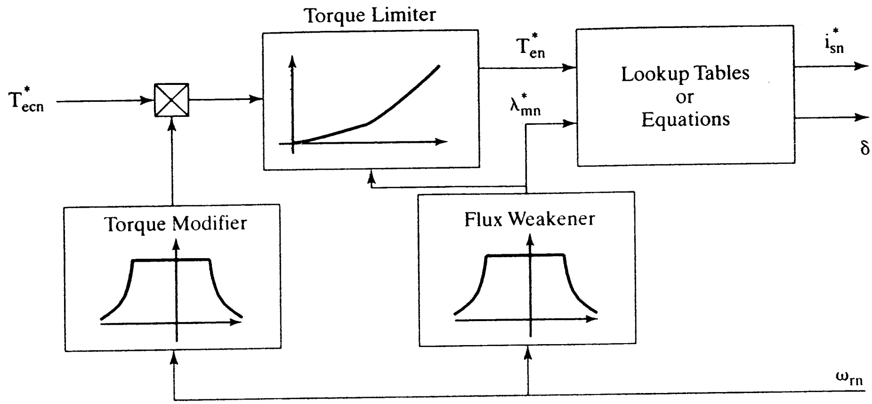


Figure 9.24 The command processor

9.6.3.3 Implementation strategy. In this section, an implementation strategy for the on-line computation of $\{i_{sn}^{*}, \delta^{*}\}$ from $\{T_{en}^{*}, \lambda_{mn}^{*}\}$ is discussed. The strategy follows this general format:

$$i_{sn}^{*} = \Omega(T_{en}^{*}, \lambda_{mn}^{*}) \tag{9.109}$$

$$\delta^{*} = \Lambda(T_{en}^{*}, \lambda_{mn}^{*}) \tag{9.110}$$

A method for realizing the functions $\Omega(.,.)$ and $\Lambda(.,.)$ based on lookup tables is presented. The method is illustrated via an example using the model parameters of a PMSM. These values are

$L_{dn} = 0.434$ p.u., $L_{qn} = 0.699$ p.u. and $R_{sn} = 0.1729$ p.u. The base values are $V_b = 97.138$ V, $I_b = 12$ A, $L_b = 0.0129$ H and $\omega_b = 628.6$ rad/sec.

Lookup Tables Realization Equations (9.109) and (9.110) can be realized by using separate three-dimensional lookup tables. These lookup tables are generated off-line by solving system equations numerically. Two of the independent axes of each table are assigned to T_{en}^{*} and λ_{mn}^{*}. Off the third dimension, the respective i_{sn}^{*} or δ^{*} is read. Note that the tables need only provide the data for positive torque commands. For negative torque commands, the respective angle from the table must be applied with negative sign. λ_{mn}^{*} is limited to values between 0 and 1, since the only interest is in weakening the mutual flux linkage. The numerical solution to the system variables for the following range of the normalized mutual-flux-linkage command:

$$0.2 \leq \lambda_{mn}^{*} \leq 1 \text{ (p.u.)}$$

and for the motor parameters given earlier, yields the viable ranges for the other three variables as

$$0 \leq T_{en}^{*} \leq 2.44 \text{ (p.u.)}$$

$$0 \leq i_{sn} \leq 3.3 \text{ (p.u.)}$$

$$1.57 \leq \delta \leq 3.14 \text{ (rad)}$$

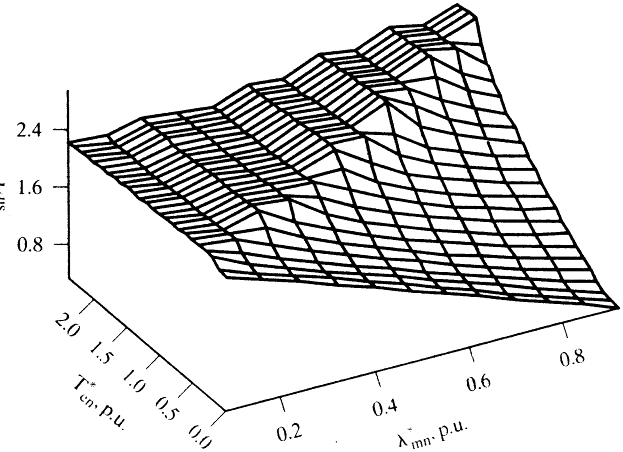


Figure 9.25 Current command as a function of torque and mutual-flux-linkage commands

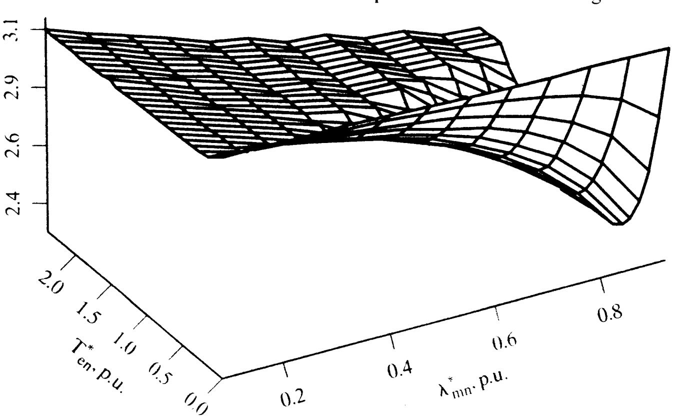


Figure 9.26 Angle command as a function of the torque and mutual-flux-linkage commands

Figures 9.25 and 9.26 depict the three-dimensional tables for this system. The variables are digitized under the following constraints,

$$T^*_{en}\text{: 9 bits; } \lambda^*_{mn}\text{: 7 bits; } i^*_{sn} \text{ and } \delta^*\text{: 8 bits each}$$

Once the torque command reaches its maximum permissible value, the current and angle commands are frozen. The fact that each graph has two distinct areas is due to this limitation. In each graph, the border between these two surfaces appears to be a straight line, which can be attributed to the fact that the T^*_{enm}-vs.-λ^*_{mn} relationship is nearly linear. The T^*_{enm}-vs.-λ^*_{mn} relationship suggests that, for the given parameters, the relationship can adequately be approximated with a first- or at most a second-order polynomial. Thus, either of the following two polynomials can be used for on-line calculation of T^*_{enm} from λ^*_{mn} in this example.

$$T^*_{enm} = 0.21\lambda^{*2}_{mn} + 2.23\lambda^*_{mn} + .0059$$

or, more simply,

$$T^*_{enm} = 2.44\lambda^*_{mn}$$

The polynomials are both derived by using the least-squares-fit method.

9.6.3.4 System performance. Utilizing these tables results in the accurate enforcement of the commanded torque and the commanded mutual flux linkage. As a result, the voltage requirement is closely maintained at a maximum level of 1 p.u. Figure 9.27 shows the simulation results for a speed-control system that utilizes the two tables for a speed-controlled operation with +3-p.u. and −3-p.u. speed commands. The dotted lines represent commanded values, the solid lines actual values. The PWM is operating at 20 kHz, and the current error is amplified by a factor of 200. The stator resistance is taken into account in the simulation; therefore, the mutual flux-weakening is initiated at a speed of .53 p.u. instead of 1 p.u. As a result, the voltage requirement is successfully limited to 1 p.u. The limited bandwidth of the current controller decreases the ability of the system to enforce desired currents as speed increases. As a consequence, the mutual-flux-linkage error increases with speed. Note that memory chips are required to store the relatively large look-up tables. Each table has 2^{16} entries. Therefore, each table requires 64 kilobytes of storage capacity.

The table-look-up approach provides a high level of accuracy in terms of enforcing the torque and mutual-flux-linkage commands. Utilizing the look-up-table method results in meeting the specified voltage requirements of the drive system, which is one of the key factors in the drive's performance. The system is also relatively fast: implementing it mainly involves reading data from a memory chip, plus the performance of minimal calculations. However, implementing this system requires a considerable amount of digital memory. For the example considered, 128 kilobytes of ROM is required to store the tables.

9.6.3.5 Parameter sensitivity. Parameter variations are a major source of error in model-based motor controllers, such as the one discussed. The effects of the variations of three of the PMSM parameters (namely, stator resistance, q axis inductance, and rotor flux linkages) on the input-voltage requirement of the simulated example are briefly discussed in this section.

(i) Stator Resistance The stator resistance can vary in the range of approximately 1 to 2 times its nominal value. As the rotor resistance increases, the input-voltage requirement also increases. Figure 9.28 shows the effect of a stator resistance increase of 100% on the voltage requirements of the simulated system. If maintaining a 1-p.u. input voltage is the main objective, the solution is to initiate the mutual-flux-weakening operation at lower speeds. Note that, as the mutual flux linkage is further weakened, the maximum permissible torque also decreases. This, in turn, increases the response time of the motor.

(ii) Rotor Flux Linkage The rotor flux linkage of a PMSM can decrease as much as 20 percent depending on the kind of magnet used. Figure 9.29 shows the effect of a 20-percent reduction of λ_{af} on the simulated example. It can be seen that the overall voltage requirement is higher than the case when the rotor flux linkage is at its

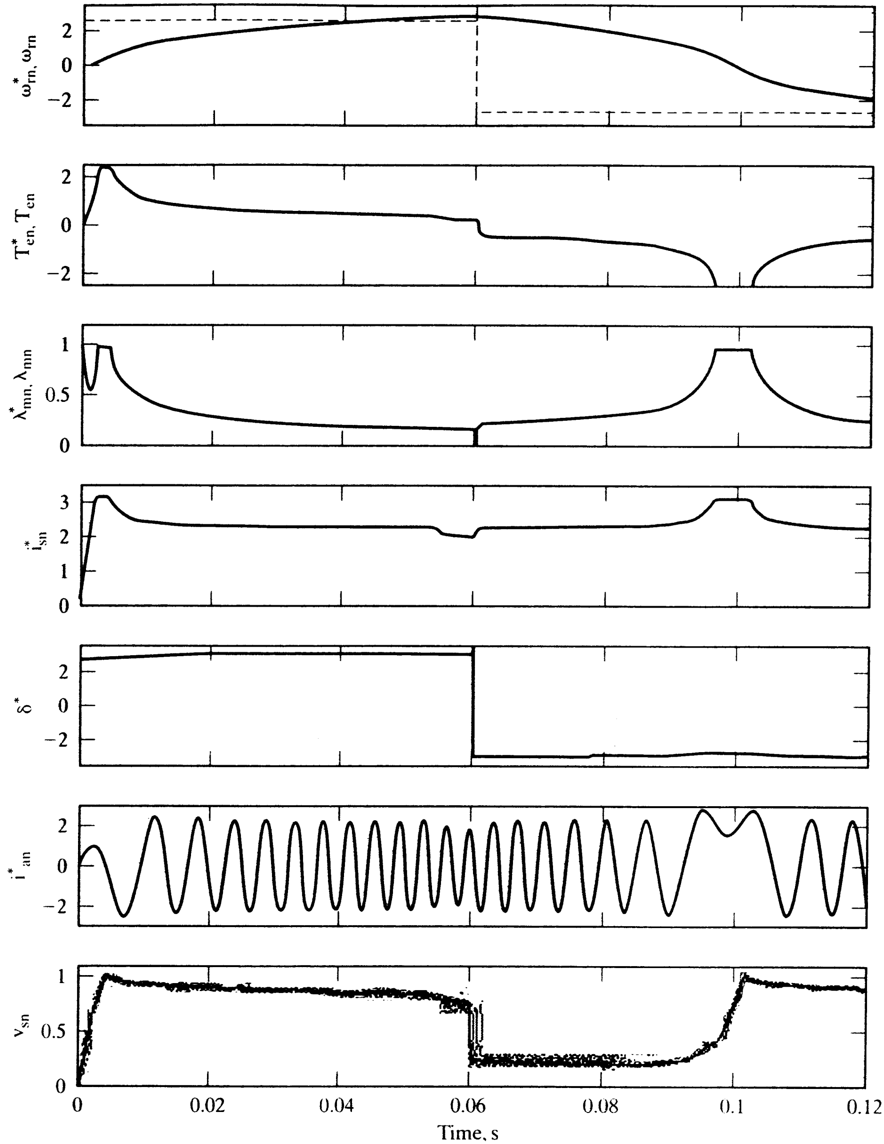
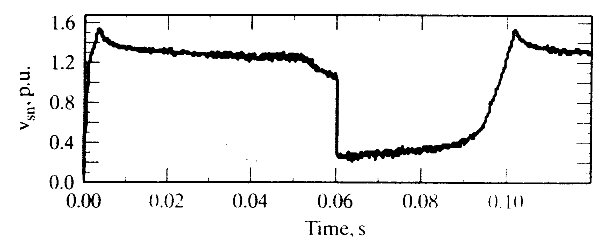


Figure 9.27 Simulation results for a ±3-p.u.-step speed command



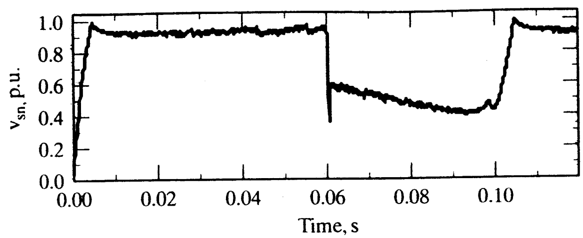


Figure 9.29 Input-stator phasor-voltage requirement when λ_{af} decreases by 20% from its nominal value

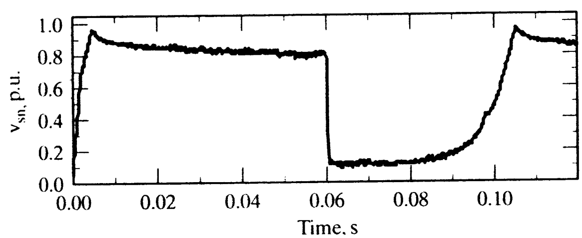


Figure 9.30 Input-voltage requirement for a 20-percent reduction in L_q

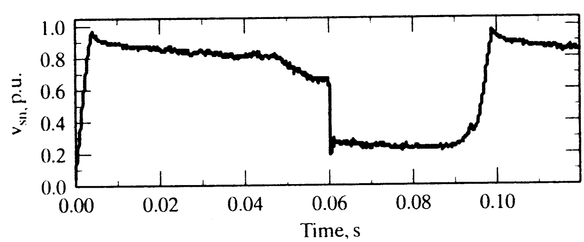


Figure 9.31 Input-voltage requirement for a 10-percent increase in L_q

nominal value. Under identical torque requirements, as the rotor flux linkage decreases, the current requirements increase and, therefore, the input-voltage requirements increase.

(iii) *q* Axis Self-Inductance The q axis self-inductance, L_q, can vary in the range of approximately from 0.8 to 1.1 times its nominal value. Figures 9.30 and 9.31 show the effect of L_q variations on the input-voltage requirement. The increase of L_q over its nominal value results in higher voltage requirements. The voltage requirement exceeds the 1-p.u. limit at some points. This indicates that a good reserve margin in the dc-link voltage has to be given to handle the parameter variations that are bound to affect the performance. The extreme cases need to be taken into account in the design process. Alternatively, estimating the individual parameters directly and using them in the controller can compensate for the effects of parameter sensitivity.

9.7 SPEED-CONTROLLER DESIGN

The design of the speed-controller is important from the point of view of imparting desired transient and steady-state characteristics to the speed-controlled PMSM drive system. A proportional-plus-integral controller is sufficient for many industrial applications; hence, it is considered in this section. Selection of the gain and time constants of such a controller by using the symmetric-optimum principle is straightforward if the d axis stator current is assumed to be zero. In the presence of a d axis stator current, the *d* and *q* current channels are cross-coupled, and the model is nonlinear, as a result of the torque term.

Under the assumption that $i^r_{ds} = 0$, the system becomes linear and resembles that of a separately-excited dc motor with constant excitation. From then on, the block-diagram derivation, current-loop approximation, speed-loop approximation, and derivation of the speed-controller by using symmetric optimum are identical to those for a dc or vector-controlled induction-motor-drive speed-controller design.

9.7.1 Block-Diagram Derivation

The motor q axis voltage equation with the d axis current being zero becomes

$$v^r_{qs} = (R_s + L_q p)i^r_{qs} + \omega_r \lambda_{af} \tag{9.111}$$

and the electromechanical equation is

$$\frac{P}{2}(T_e - T_l) = Jp\omega_r + B_1\omega_r \tag{9.112}$$

where the electromagnetic torque is given by

$$T_e = \frac{3}{2} \cdot \frac{P}{2} \lambda_{af} i^r_{qs} \tag{9.113}$$

and, if the load is assumed to be frictional, then

$$T_l = B_l \omega_m \tag{9.114}$$

which, upon substitution, gives the electromechanical equation as

$$(Jp + B_t)\omega_r = \left\{\frac{3}{2}\left(\frac{P}{2}\right)^2 \cdot \lambda_{af}\right\} i^r_{qs} = K_t \cdot i^r_{qs} \tag{9.115}$$

where

$$B_t = \frac{P}{2}B_l + B_1 \tag{9.116}$$

$$K_t = \frac{3}{2}\left(\frac{P}{2}\right)^2 \cdot \lambda_{af} \tag{9.117}$$

The equations (9.111) and (9.115), when combined into a block diagram with the current- and speed-feedback loops added, are shown in Figure 9.32.

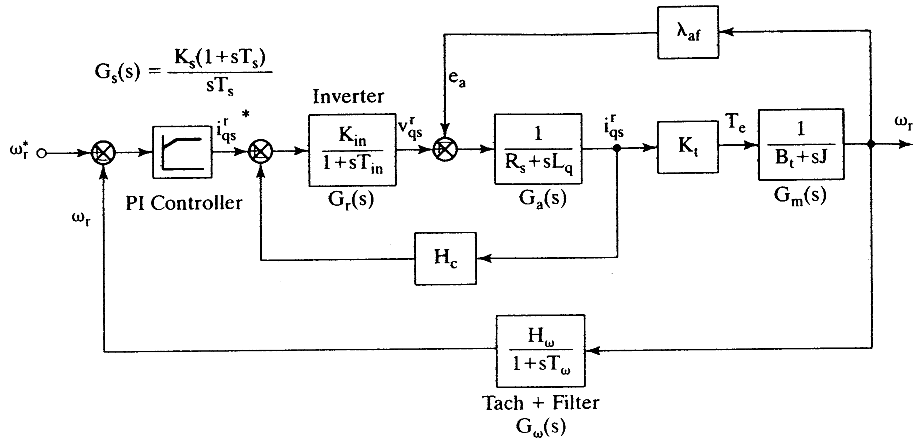


Figure 9.32 Block diagram of the speed-controlled PMSM drive

The inverter is modeled as a gain with a time lag by

$$G_r(s) = \frac{K_{in}}{1 + sT_{in}} \tag{9.118}$$

where

$$\left.\begin{aligned} K_{in} &= 0.65\frac{V_{dc}}{V_{cm}} \\ T_{in} &= \frac{1}{2f_c} \end{aligned}\right\} \tag{9.119}$$

where V_{dc} is the dc-link voltage input to the inverter, V_{cm} is the maximum control voltage, and f_c is the switching (carrier) frequency of the inverter.

The induced emf due to rotor flux linkages, e_a, is

$$e_a = \lambda_{af}\omega_r \text{ (V)} \tag{9.120}$$

9.7.2 Current Loop

This induced-emf loop crosses the q axis current loop, and it could be simplified by moving the pick-off point for the induced-emf loop from speed to current output point. This gives the current-loop transfer function from Figure 9.33 as

$$\frac{i_{qs}^r(s)}{i_{qs}^r{}^*(s)} = \frac{K_{in}K_a(1 + sT_m)}{H_cK_aK_{in}(1 + sT_m) + (1 + sT_{in})\{K_aK_b + (1 + sT_a)(1 + sT_m)\}} \tag{9.121}$$

where

$$K_a = \frac{1}{R_s};\ T_a = \frac{L_q}{R_s};\ K_m = \frac{1}{B_t};\ T_m = \frac{J}{B_t};\ K_b = K_tK_m\lambda_{af} \tag{9.122}$$

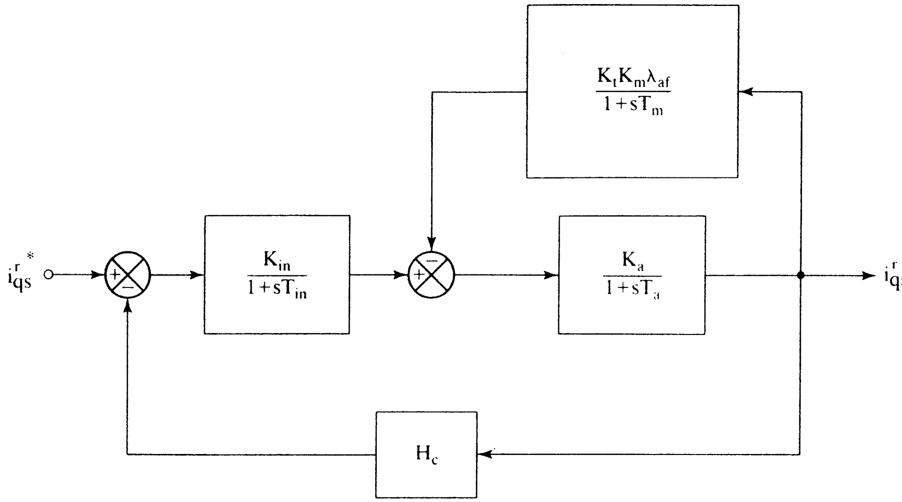


Figure 9.33 Current controller

The following approximations are valid near the vicinity of crossover frequency:

$$1 + sT_r \cong 1 \tag{9.123}$$

$$1 + sT_m \cong sT_m \tag{9.124}$$

$$(1 + sT_a)(1 + sT_{in}) \cong 1 + s(T_a + T_{in}) \cong 1 + sT_{ar} \tag{9.125}$$

where

$$T_{ar} = T_a + T_{in} \tag{9.126}$$

With this, the current-loop transfer function is approximated as

$$\frac{i_{qs}^r(s)}{i_{qs}^r*(s)} \cong \frac{(K_a K_{in} T_m)s}{K_a K_b + (T_m + K_a K_{in} T_m H_c)s + (T_m T_{ar})s^2} \cong \left(\frac{T_m K_{in}}{K_b}\right)\frac{s}{(1 + sT_1)(1 + sT_2)} \tag{9.127}$$

It is found that $T_1 < T_2 < T_m$; hence, on further approximation, $(1 + sT_2) \cong sT_2$. The approximate current-loop transfer function is then given by

$$\frac{i_{qs}^r(s)}{i_{qs}^r*(s)} \cong \frac{K_i}{(1 + sT_i)} \tag{9.128}$$

where

$$K_i = \frac{T_m K_{in}}{T_2 K_b} \tag{9.129}$$

$$T_i = T_1 \tag{9.130}$$

This simplified current-loop transfer function is substituted in the design of the speed controller as follows.

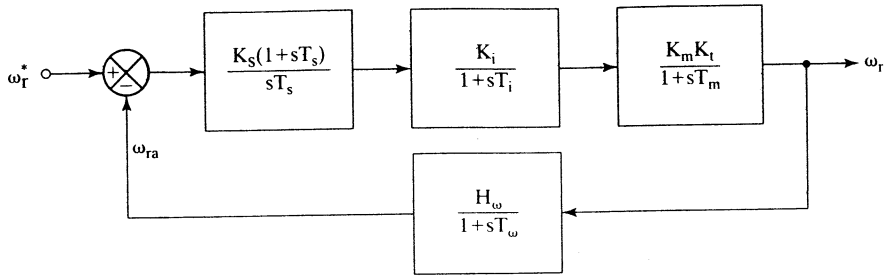


Figure 9.34 Simplified speed-control loop

9.7.3 Speed-Controller

The speed loop with the simplified current loop is shown in Figure 9.34.

Near the vicinity of the crossover frequency, the following approximations are valid:

$$(1 + sT_m) \cong sT_m \tag{9.131}$$

$$(1 + sT_i)(1 + sT_\omega) \cong 1 + sT_{\omega i} \tag{9.132}$$

$$1 + sT_\omega \cong 1 \tag{9.133}$$

where

$$T_{\omega i} = T_\omega + T_i \tag{9.134}$$

The speed-loop transfer function, with these approximations, is given by

$$GH(s) \cong \frac{K_i K_m K_t H_\omega}{T_m} \cdot \frac{K_s}{T_s} \cdot \frac{(1 + sT_s)}{s^2(1 + sT_{\omega i})} \tag{9.135}$$

from which the closed-loop speed transfer function is obtained as

$$\frac{\omega_r(s)}{\omega_r^*(s)} \cong \frac{1}{H_\omega}\left\{\frac{K_g \frac{K_s}{T_s}(1 + sT_s)}{s^3 T_{\omega i} + s^2 + K_g \frac{K_s}{T_s}(1 + sT_s)}\right\} \tag{9.136}$$

where

$$K_g = \frac{K_i K_m K_t H_\omega}{T_m} \tag{9.137}$$

Equating this transfer function to a symmetric-optimum function with a damping ratio of 0.707 gives the closed-loop-speed transfer function as

$$\frac{\omega_r(s)}{\omega_r^*(s)} \cong \frac{1}{H_\omega} \cdot \frac{(1 + sT_s)}{1 + (T_s)s + \left(\frac{3}{8}T_s^2\right)s^2 + \left(\frac{1}{16}T_s^3\right)s^3} \tag{9.138}$$

Equating the coefficients of equations (9.136) and (9.138) and solving for the constants yields the time and gain constants of the speed controller as

$$T_s = 6T_{\omega i} \tag{9.139}$$

$$K_s = \frac{4}{9K_gT_{\omega i}} \tag{9.140}$$

Hence, the proportional gain, K_{ps}, and integral gain, K_{is}, of the speed controller are derived as

$$K_{ps} = K_s = \frac{4}{9K_gT_{\omega i}} \tag{9.141}$$

$$K_{is} = \frac{K_s}{T_s} = \frac{1}{27K_gT_{\omega i}^2} \tag{9.142}$$

The validity of various approximations is verified through a worked example.

Example 9.2

The PMSM drive system parameters are as follows:

$R_s = 1.4\ \Omega, L_d = 0.0056$ H, $L_q = 0.009$ H, $\lambda_{af} = 0.1546$ Wb−Turn, $B_t = 0.01$ N·m/rad/sec, $J = 0.006$ kg − m², $P = 6$, $f_c = 2$ kHz, $V_{cm} = 10$ V, $H_\omega = 0.05$ V/V, $H_c = 0.8$ V/A, $V_{dc} = 285$ V.

Design a symmetric-optimum-based speed-controller, and verify the validity of assumptions made in its derivation. The damping ratio required is 0.707.

Solution

Inverter: Gain, $K_{in} = 0.65\dfrac{V_{dc}}{V_{cm}} = 18.525$ V/V

Time constant, $T_{in} = \dfrac{1}{2f_c} = 0.00025$ s

$$G_r(s) = \frac{K_{in}}{1 + sT_{in}} = \frac{18.525}{(1 + 0.00025\ s)}$$

Motor (electrical): Gain, $K_a = \dfrac{1}{R_s} = 0.7143$; Time constant, $T_a = \dfrac{L_q}{R_s} = 0.0064$ s

$$G_a(s) = \frac{K_a}{1 + sT_a} = \frac{0.7143}{(1 + 0.0064\ s)}$$

Induced emf loop: Torque constant, $K_t = \dfrac{3}{2}\left(\dfrac{P}{2}\right)^2 \cdot \lambda_{af} = 2.087$ N·m/A

Mechanical gain, $K_m = \dfrac{1}{B_t} = 100$ rad/s/Nm

$$G_b(s) = \frac{K_tK_m\lambda_{af}}{(1 + sT_m)} = \frac{32.26}{(1 + 0.6s)}$$

where the mechanical time constant is

$$T_m = \frac{J}{B_t} = 0.6\ s$$

Motor (mechanical): $G_m(s) = \dfrac{K_m K_t}{(1 + sT_m)} = \dfrac{208.7}{(1 + 0.6\,s)}$

Equivalent electrical time constants of the motor: Solve for the roots of $as^2 + bs + c = 0$, where

$$a = T_m T_{ar}$$
$$b = T_m + K_a K_{in} T_m H_c$$
$$c = K_a K_b$$

where

$$K_b = K_t K_m \lambda_{af} = 32.26$$

Then the inverse of the roots will give T_1 and T_2 as

$$T_1 = 0.0005775\ (\text{sec})$$
$$T_2 = 0.301\ (\text{sec})$$

Simplified current-loop transfer function: $G_{is}(s) = \dfrac{K_i}{1 + sT_i}$

$$T_i = T_1 = 0.0005775\ (\text{s})$$
$$K_i = \frac{T_m K_r}{T_2 K_b} = 1.1443$$

Exact current-loop transfer function: $G_i(s) = \dfrac{G_r(s) \cdot G_a(s)/[1 + G_a(s) \cdot G_b(s)]}{1 + H_c \cdot G_a(s) \cdot G(s)/[1 + G_a(s) \cdot G_b(s)]}$

Speed controller:

$$K_g = K_i K_m K_t \frac{H_\omega}{T_m} = 19.90$$
$$T_{\omega i} = T_\omega + T_i = 0.0025775\ (\text{s})$$
$$T_s = 6T_{\omega i} = 0.0155\ (\text{s})$$
$$K_s = \frac{4}{9K_g T_{\omega i}} = 8.6638$$

Simplified speed-loop transfer function: $G_{ss}(s) \cong \dfrac{1}{H_\omega}\left\{\dfrac{K_g \dfrac{K_s}{T_s}(1 + sT_s)}{s^3(T_{\omega i}) + s^2 + K_g \dfrac{K_s}{T_s}(1 + sT_s)}\right\}$

Exact speed-loop transfer function: $G_{se}(s) = \dfrac{G_m(s) \cdot G_i(s) \cdot G_s(s)}{1 + G_\omega(s) \cdot G_m(s) \cdot G_i(s) \cdot G_s(s)}$

where

$$G_s(s) = \frac{K_s}{T_s} \cdot \frac{(1 + sT_s)}{s} = (560.2)\frac{(1 + 0.0155\,s)}{2}$$
$$G_\omega(s) = \frac{H_\omega}{1 + sT_\omega} = \frac{0.05}{(1 + 0.002\,s)}$$

Smoothing: It is achieved by canceling the zero, $(1 + sT_s)$, with a pole inserted in series with the speed reference. Note that this leaves the final speed-loop transfer function with only poles. The smoothing can also be thought of as the soft-start controller employed in almost all of the drive systems in practice.

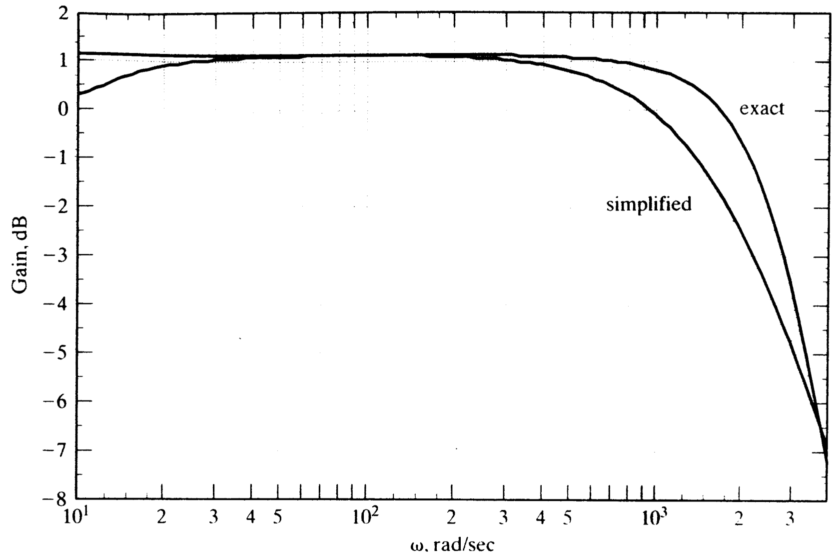


Figure 9.35 Exact and simplified current-loop-gain plots

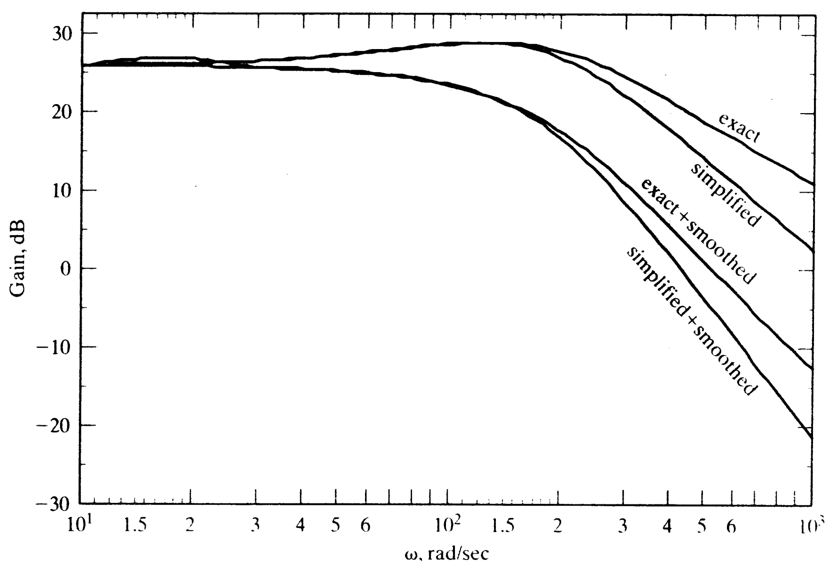


Figure 9.36 Gain plots of various speed-loop transfer functions

All the transfer-function gains and phases are plotted in Figures 9.35, 9.36, and 9.37. In the frequency regions of interest, note that the approximations hold good both in magnitude and in phase, in spite of the reduction of the fifth-order system to an equivalent third-order in the case of the speed loop and of a third to a first in the current loop.

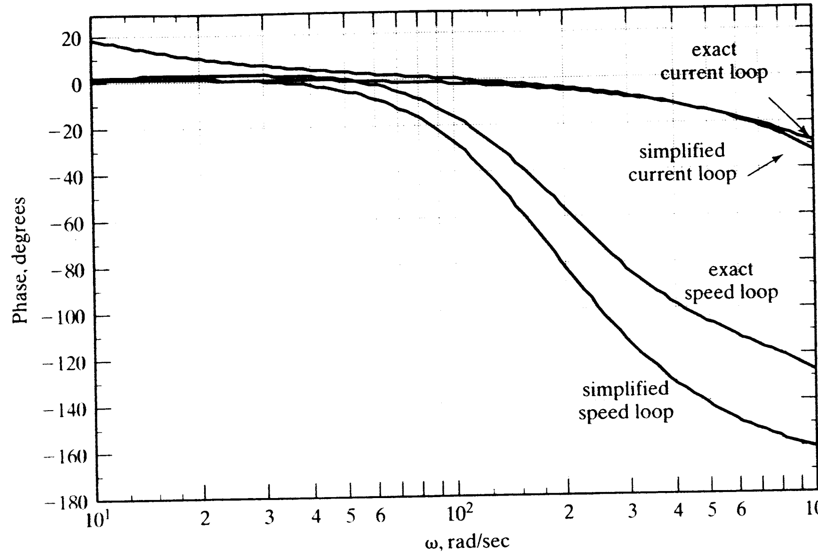


Figure 9.37 Phase plots of exact and simplified current and speed-loop transfer functions

9.8 SENSORLESS CONTROL

The PMSM drive requires two current sensors and an absolute-rotor-position sensor for the implementation of any control strategy discussed in the previous section. The rotor position is sensed with an optical encoder or a resolver for high-performance applications. The position sensors compare to the cost of the low-power motor, thus making the total system cost very noncompetitive compared to other types of motor drives. As for the current sensors, they are not as expensive as the rotor-position sensor; note that other types of drives also require their use in feedback control. Hence, the control and operation of PMSM drive without a rotor-position sensor would enhance its applicability to many cost-sensitive applications and provide a back-up control in sensor-based drives during sensor failures. One method of rotor-position sensorless control strategy is discussed in this section.

The basis for this control strategy is that the error between measured and calculated currents from the machine model gives the difference between the assumed rotor speed and the actual rotor speed of the motor drive. Nulling this current error results in the synchronous operation of the motor drive by estimating its rotor position accurately. The relevant algorithm for sensorless control is developed next.

The following assumptions are made to develop this control algorithm.

(i) Motor parameters and rotor PM flux are constant.

(ii) Induced emfs in the machine are sinusoidal.

(iii) The drive operates in the constant-torque region, and flux-weakening operation is not considered.

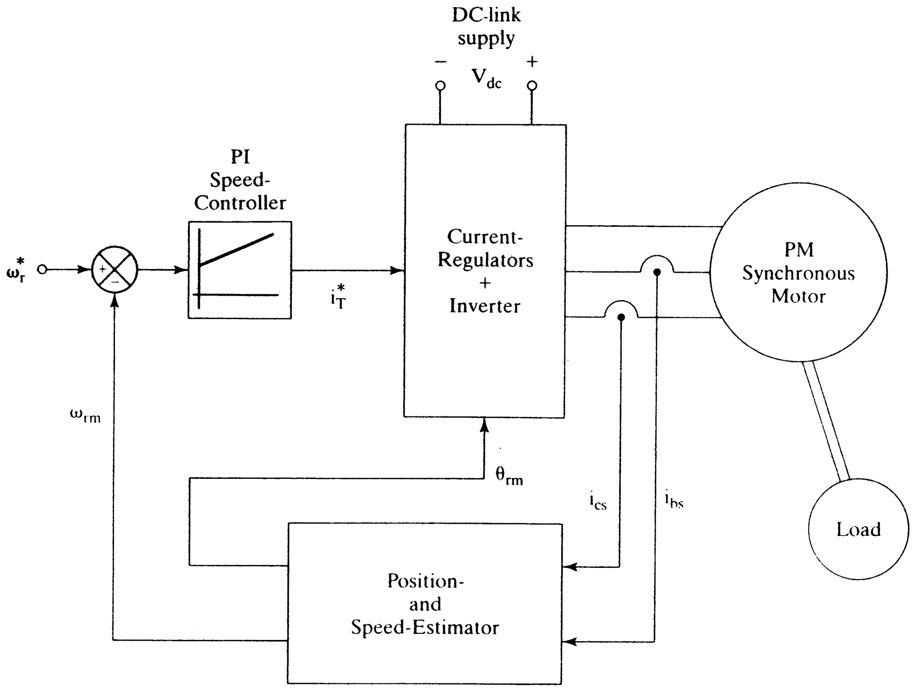


Figure 9.38 Control schematic diagram of the sensorless PMSM drive

The basic control schematic diagram is shown in Figure 9.38. Two phase currents constitute the inputs to the electrical rotor position and speed estimator. The error between the reference speed, ω_r^*, and the estimated rotor speed, ω_{rm}, is amplified and limited to provide the torque-producing component of the stator current, i_T^*, which is the q axis stator current in the rotor reference frames. The estimated rotor position, together with i_T^*, provides the stator current commands, which are enforced by a three-phase inverter feeding the PMSM. The position and speed estimator is derived from the machine equations.

Consider that the machine is running at a speed ω_r, whereas the model starts with an assumed rotor speed ω_{rm}. This is shown in the phasor diagram given in Figure 9.39. The assumed rotor position θ_{rm} lags behind the actual rotor position θ_r by $\delta\theta$ radians. They are related to the actual and assumed or model speed as follows:

$$\theta_r = \int \omega_r dt \tag{9.143}$$

$$\theta_{rm} = \int \omega_{rm} dt \tag{9.144}$$

$$\delta\theta = \theta_r - \theta_{rm} = \int (\omega_r - \omega_{rm}) dt \tag{9.145}$$

The machine model is utilized to compute the stator currents; note that it is carried out in a reference frame at an assumed rotor speed. That implies the reference frames are α and β axes and not d and q axes, which are the usual rotor reference

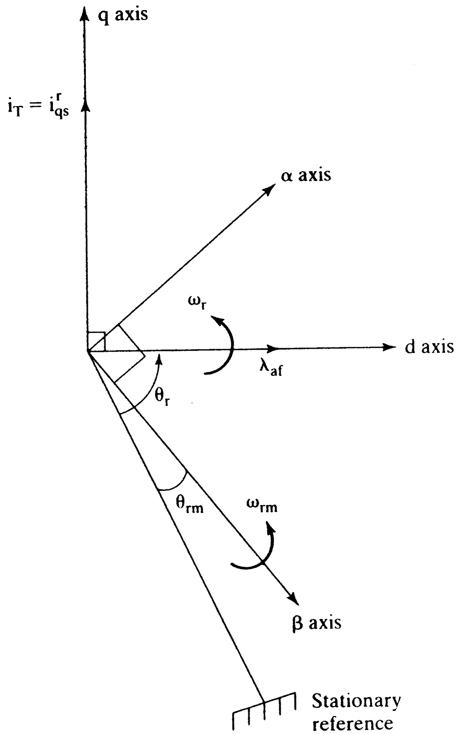


Figure 9.39 Phasor diagram corresponding to an error between the actual and assumed rotor position

frames. Therefore, the machine equations in the assumed rotor-speed reference frames are

$$\begin{bmatrix} pi_{\alpha m} \\ pi_{\beta m} \end{bmatrix} = \begin{bmatrix} -\dfrac{R_s}{L_q} & -\omega_{rm}\dfrac{L_d}{L_q} \\ \omega_{rm}\dfrac{L_q}{L_d} & -\dfrac{R_s}{L_q} \end{bmatrix} \begin{bmatrix} i_{\alpha m} \\ i_{\beta m} \end{bmatrix} + \begin{bmatrix} -\dfrac{\omega_{rm}\lambda_{af}}{L_q} \\ 0 \end{bmatrix} + \begin{bmatrix} \dfrac{v_\alpha}{L_q} \\ \dfrac{v_\beta}{L_d} \end{bmatrix} \tag{9.146}$$

In the model α and β reference frames, the actual machine equations are written from the *d* and *q* axes as

$$\begin{bmatrix} pi_{\alpha} \\ pi_{\beta} \end{bmatrix} = \begin{bmatrix} -\dfrac{R_s}{L_q} & -\omega_{rm}\dfrac{L_d}{L_q} \\ \omega_{rm}\dfrac{L_q}{L_d} & -\dfrac{R_s}{L_q} \end{bmatrix} \begin{bmatrix} i_{\alpha} \\ i_{\beta} \end{bmatrix} + \begin{bmatrix} -\dfrac{\omega_r\lambda_{af}}{L_q}\cos\delta\theta \\ \dfrac{\omega_r\lambda_{af}}{L_d}\sin\delta\theta \end{bmatrix} + \begin{bmatrix} \dfrac{v_\alpha}{L_q} \\ \dfrac{v_\beta}{L_d} \end{bmatrix} \tag{9.147}$$

The variables without the second subscript indicate that they are actual machine variables; the machine model (or estimated) variables end with subscript *m*. The actual machine equations are derived on the understanding that α–β axes are the considered reference axes and hence the rotor flux linkages have components on them from the *d* axis given as a function of the error in rotor position, $\delta\theta$. It is assumed that the entire rotor field is aligned on the d axis.

Discretize the two sets of model and actual current equations, respectively, as

$$\begin{bmatrix} i_{\alpha m}(kT) \\ i_{\beta m}(kT) \end{bmatrix} = \begin{bmatrix} i_{\alpha m}(\overline{k-1T}) \\ i_{\beta m}(\overline{k-1T}) \end{bmatrix} + \begin{bmatrix} pi_{\alpha m}(\overline{k-1T}) \\ pi_{\beta m}(\overline{k-1T}) \end{bmatrix} T \tag{9.148}$$

$$\begin{bmatrix} i_{\alpha}(kT) \\ i_{\beta}(kT) \end{bmatrix} = \begin{bmatrix} i_{\alpha}(\overline{k-1T}) \\ i_{\beta}(\overline{k-1T}) \end{bmatrix} + \begin{bmatrix} pi_{\alpha}(\overline{k-1T}) \\ pi_{\beta}(\overline{k-1T}) \end{bmatrix} T \tag{9.149}$$

where T is the sampling time and *k* is the present sampling instant. Substituting for the derivative terms in terms of the currents, induced emfs, and input voltages from equations (9.146) and (9.147) into (9.148) and (9.149) and finding their respective current errors yields the following:

$$\begin{bmatrix} \delta i_{\alpha}(kT) \\ \delta i_{\beta}(kT) \end{bmatrix} = \begin{bmatrix} i_{\alpha}(kT) - i_{\alpha m}(kT) \\ i_{\beta}(kT) - i_{\beta m}(kT) \end{bmatrix} = T \begin{bmatrix} -\dfrac{\lambda_{af}}{L_q}(\omega_r \cos\delta\theta - \omega_{rm}) \\ \dfrac{\omega_r \lambda_{af}}{L_d}\sin\delta\theta \end{bmatrix} \tag{9.150}$$

A number of assumptions have been made to derive (9.150): (i) The difference between the model and actual currents, when multiplied by T, becomes negligible; (ii) The sampling time is very small compared to the electrical and mechanical time constants of the drive system.

If $\delta\theta$ is small, then the following approximations are valid for use in interpreting the above results:

$$\sin\delta\theta \cong \delta\theta \tag{9.151}$$

$$\cos\delta\theta \cong 1 \tag{9.152}$$

Hence, substituting these into error currents results in

$$\delta i_{\alpha}(kT) = -\frac{\lambda_{af}}{L_q} T(-\omega_{rm} + \omega_r) \tag{9.153}$$

$$\delta i_{\beta}(kT) = \omega_r \frac{\lambda_{af}}{L_d} \delta\theta \tag{9.154}$$

from which the actual rotor speed is obtained as

$$\omega_r = -\frac{L_q}{\lambda_{af}} \frac{1}{T} \delta i_{\alpha}(kT) + \omega_{rm} \tag{9.155}$$

and the error in estimated rotor position is

$$\delta\theta = \frac{L_d}{\lambda_{af}} \frac{1}{T} \frac{\delta i_{\beta}(kT)}{\omega_r} \tag{9.156}$$

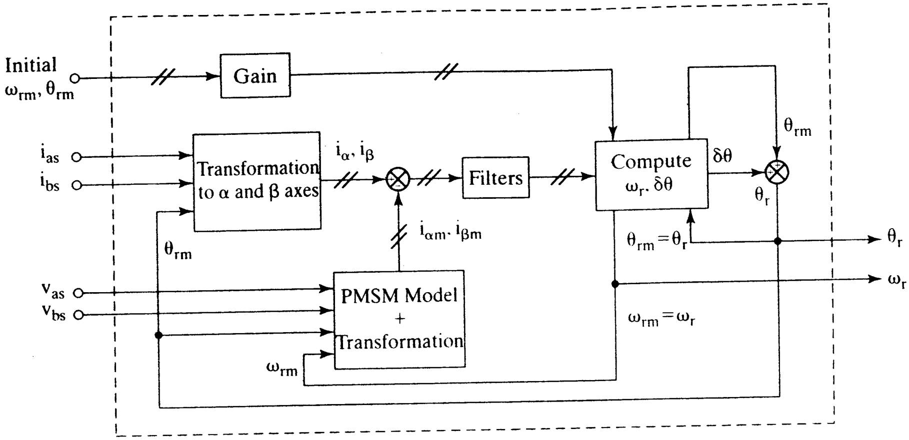


Figure 9.40 Block-diagram realization of electrical rotor position and speed estimation

Substitution for the rotor speed in the error rotor-position equation gives $\delta\theta$ as

$$\delta\theta = \frac{\left(\frac{L_d}{T\lambda_{af}}\right)\delta i_\beta(kT)}{\left[\omega_{rm} - \frac{L_q}{T\lambda_{af}}\delta i_\alpha(kT)\right]} \tag{9.157}$$

Note that $\delta\theta$, ω_{rm}, and ω_r are for the sampling instant of kT also. They have to be evaluated for each sampling instant to follow the rotor position closely. The rotor position then is

$$\theta_r = \theta_{rm} + \delta\theta \tag{9.158}$$

which becomes the estimate for θ_{rm} in the next sampling interval for feeding into the controller to compute stator-current commands. Filters are required to smooth ripples in the current errors due to PWM voltages fed to the machine. The realization of the position and speed estimator is shown in block-diagram form in Figure 9.40.

Starting from standstill is achieved by bringing the rotor to a particular position by energizing the stator phases accordingly. Alternatively, a sequence of pulse patterns is issued to the inverter, which would enable the motor to achieve a significant but small rotor speed to start the estimation. During this time, the estimator is kept inactive; it is brought into the control process by switching off the starting process. The input of initial values of speed and rotor position serves to smooth the transition process from initial starting to estimation for continued operation and control. Low-speed operation continues to be a challenge with this technique.

Several other methods exist to estimate the rotor position. For one, the induced emf of the stator phases could be estimated from the measured stator currents and voltages. This method has the problem of finding the position at zero speed; there are no induced emfs at that point, and at low speeds they would be difficult to estimate accurately because of small magnitude. Hence, this method has to incorporate an initial-starting process, as discussed above.

Ideally, rotor position can be obtained from stator self-inductances. The reluctance variation between, say, q and d axes in the machine occurs when the magnets have a pole arc less than 180 electrical degrees, regardless of the methods of magnet placement in the rotor. This variation in the reluctance can be measured by noninvasive measurement techniques, and then they can be used to extract the rotor position information. This reluctance variation is prominent in machines with high saliency. Even the surface-magnet machines exhibit a 10% variation in their reluctance between the d and q axes. References contain research papers discussing alternative methods of estimating rotor position.

9.9 PARAMETER SENSITIVITY

Temperature variation changes the stator resistance and flux remanence in the permanent magnets. The loss of magnetism for a 100°C rise in temperature in ceramic, neodymium, and samarium–cobalt magnets is 19%, 11%, and 4%, respectively, from their nominal values. The effect due to the loss of magnetism with temperature variations is predominant compared to the effect of stator-resistance variations on the performance of the drive system. Further, the stator-resistance sensitivity is overcome in current-regulated drives by the nature of closed-loop control. That the current-control has no impact on the drive system is due to temperature sensitivity of the magnets. A closed-loop speed-controlled drive system will minimize the effects due to temperature sensitivity of the magnets. To have an understanding of this operation, consider that the drive system is in steady state and that the magnet flux density has decreased in a step fashion (not usually the case, but taken up here as an extreme case for illustration). The torque will decrease instantaneously, because the current-magnitude command is a constant in steady state. With the decrease in the torque, the rotor will slow down, resulting in higher speed error, higher torque command, and hence higher current command. The torque then will rise and hence the rotor speed, and this cycle of events will go on, depending on the dynamics of the system, until the system reaches steady state again. In the wake of such a disturbance in the magnet flux linkages, the drive system will encounter electromagnetic torque oscillations as explained, and that may not be very desirable in high-performance drives.

Saturation usually will affect the quadrature-axis inductance rather than the direct-axis inductance. It is due to the fact that the direct axis, with its magnets, presents a very high-reluctance path; the magnets have a relative permeability comparable to that of air. In the quadrature axis, the reluctance is lower, because most of the flux path is through the iron.

By denoting the temperature-variation effects by α and the saturation effects by β, the relationship of the torque to its reference and of the mutual flux linkage to its reference are derived with the speed loop open for the PMSM drive. It is further assumed that the drive has inner current loops and that the stator currents equal their references.

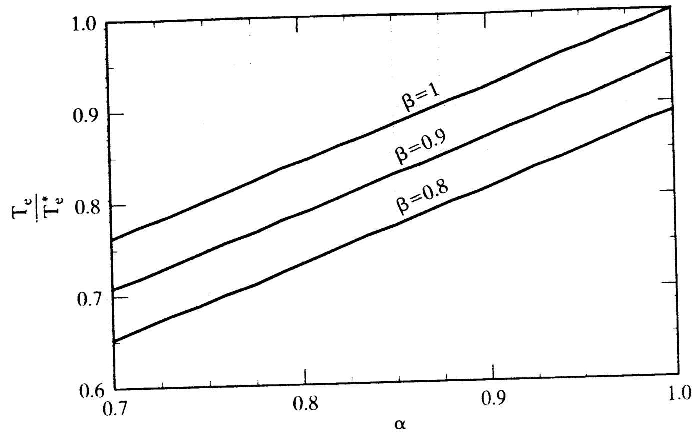


Figure 9.41 Ratio of electromagnetic torque to its command for various saturation levels, with speed loop open

9.9.1 Ratio of Torque to Its Reference

Ratio of torque to its reference is derived as

$$\frac{T_e}{T_e^*} = \frac{\alpha\lambda_{af}^* i_{qs}^{r*} + (L_d - \beta L_q) i_{ds}^{r*} i_{qs}^{r*}}{\lambda_{af}^* i_{qs}^{r*} + (L_d - L_q) i_{ds}^{r*} i_{qs}^{r*}} = \frac{\alpha\lambda_{af}^* + L_d(1 - \beta\rho) i_{ds}^{r*}}{\lambda_{af}^* + L_d(1 - \rho) i_{ds}^{r*}} \tag{9.159}$$

Note that the ratio of torque to its reference is independent of the q axis stator current. By dividing the numerator and denominator by the base flux linkages, $L_b I_b$, the torque-to-reference is derived as

$$\frac{T_e}{T_e^*} = \frac{\alpha\lambda_{afn} + (1 - \beta\rho) L_{dn} i_{dsn}^r}{\lambda_{afn} + (1 - \rho) L_{dn} i_{dsn}^r} \tag{9.160}$$

where the normalized rotor flux linkage is given as

$$\lambda_{afn} = \frac{\lambda_{af}}{L_b I_b} = \frac{\lambda_{af}}{\lambda_b}, \text{ p.u.} \tag{9.161}$$

Note that

$$i_{dsn}^r = (i_{dsn}^r)^* \tag{9.162}$$

For the same machine details used in the previous illustrations, the ratio of torque to its reference, plotted against α, is shown in Figure 9.41 for various values of β ranging from 80% to 100% of the nominal value of L_q. Note that $i_{dn}^r = -1$ p.u., $\rho = 1.607$, and $L_{dn} = 0.435$ p.u.. α is varied from 0.7 to 1. Lower values of α indicate increased rotor temperature; the value of 1 corresponds to the operation at ambient temperature. Higher rotor temperature reduces the output torque for the same stator current, and saturation further decreases it from the nominal value. The variation is linear, unlike in the case of the indirect vector controlled induction motor drive described in Chapter 8.

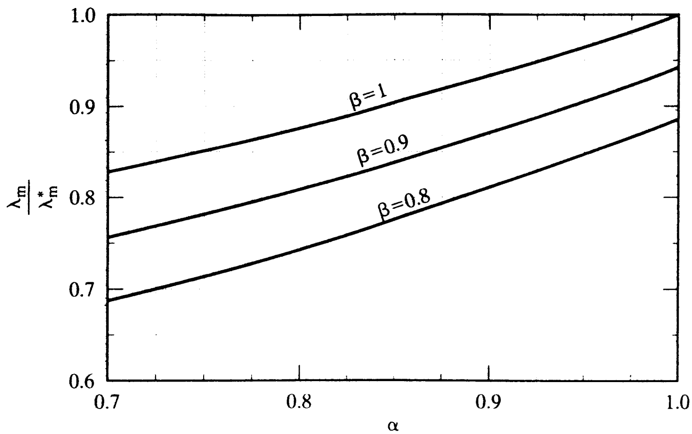


Figure 9.42 Ratio of mutual flux linkage to its reference value for various saturation levels, with speed loop open

Note that, when torque angle is set at $\pi/2$, $i^r_{dsn} = 0$, and hence the torque-to-reference equals α. Then it becomes independent of saturation. This is contrary to the performance of an indirect vector-controlled induction motor.

9.9.2 Ratio of Mutual Flux Linkage to Its Reference

The ratio of mutual flux linkage to its reference is derived as

$$\frac{\lambda_m}{\lambda_m^*} = \frac{\sqrt{(\alpha\lambda_{afn} + L_{dn}i^r_{dsn})^2 + (\beta L_{qn}i^r_{qsn})^2}}{\sqrt{(\lambda_{afn} + L_{dn}i^r_{dsn})^2 + (L_{qn}i^r_{qsn})^2}} = \frac{\sqrt{(\alpha\lambda_{afn} + L_{dn}i^r_{dsn})^2 + (\beta\rho L_{dn}i^r_{qsn})^2}}{\sqrt{(\lambda_{afn} + L_{dn}i^r_{dsn})^2 + (\rho L_{dn}i^r_{qsn})^2}} \tag{9.163}$$

where ρ is the saliency factor given by the ratio between the quadrature- and direct-axis inductances. For the same values considered in the previous illustration, the ratio of the mutual flux linkage to its reference vs. α for various β values is shown in Figure 9.42. They follow the trend of the torque-to-reference-vs.-α characteristics, except that they are not linear any more.

Temperature and saturation variations produce a nonideal torque and mutual-flux-linkages amplifier of the PMSM drive, with the consequence that the motor drive is not suitable for precision torque and speed-control applications. The changes in the parameter variations could be detected and compensated in a manner similar to the methods described for vector-controlled induction motor drives. One such method is given in the following section.

9.9.3 Parameter Compensation Through Air Gap Power Feedback Control

Air gap power is a variable that is a clear indicator of the variations in rotor flux linkages and saturation in the q axis of the PMSM. Air gap power is computed from

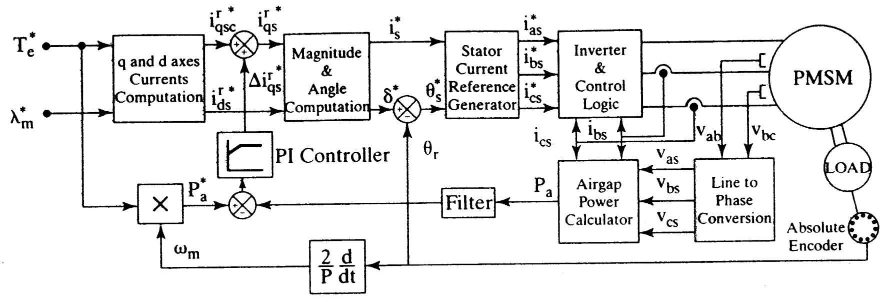


Figure 9.43 Parameter compensation of PMSM drive with air-gap-power feedback control

the input power of the machine by subtracting the stator resistance losses. If this air gap power is filtered to remove the switching ripple due to stator currents and transient magnetic energy, then it is a dc signal for a given operating point. This is denoted as

$$P_a = P_i - \frac{3}{2}\{(i^r_{qs})^2 + (i^r_{ds})^2\}R_s \tag{9.164}$$

where input power, P_i, is

$$P_i = \frac{3}{2}\{v^r_{qs}i^r_{qs} + v^r_{ds}i^r_{ds}\} = v_{as}i_{as} + v_{bs}i_{bs} + v_{cs}i_{cs} \tag{9.165}$$

The variations in the machine rotor flux linkages are embodied in the q axis voltage, and saturation effect in the form of L_q variation is contained in the d axis voltage. Hence, air gap power indicates the effects of major parameter variations, including that of R_s. The closure of air gap power feedback control requires the determination of reference air gap power.

The reference air gap power is obtained from the reference or actual speed and the torque reference in the drive system as

$$P^*_a = \frac{\omega_r T^*_e}{(P/2)} \tag{9.166}$$

The error between the reference and the computed/measured air gap power is amplified and then limited to provide a *q* axis stator-current correction signal to compensate for the parameter variations, as is shown in Figure 9.43. This signal, denoted as Δi^*_{qs}, is summed with the calculated *q* axis current, $(i^r_{qsc})^*$, to provide the compensated *q* axis current reference, $(i^r_{qs})^*$.

The working of this feedback loop is explained as follows. Assuming a reduction in the rotor flux linkage from its nominal value, it is deduced that q axis voltage will be reduced; hence, a reduction in the measured/computed air gap power, P_a, sets in. This will result in a positive and a larger $\Delta i^r_{qs}{}^*$, thus increasing the stator q axis current and so resulting in increasing air gap power to equal its reference, P^*_a. Similar reasoning would show that compensation for saturation effect also is achieved with this control strategy.

9.9.3.1 Algorithm. The correction signal is denoted as Δi_{qs}^{r*} and is given by

$$\Delta i_{qs}^{r*} = K_p(P_a^* - P_a) + K_i \int (P_a^* - P_a) \quad (9.167)$$

where K_p and K_i are the proportional and integral gains of the PI controller.

Assuming no d axis stator current, the q axis current command in rotor reference frame is obtained as

$$i_{qs}^{r*} = \frac{T_e^*}{\frac{3}{2}\frac{P}{2}\lambda_{af}^*} \quad (9.168)$$

The reference torque component of the current is augmented by the correction signal as

$$i_T^* = i_{qs}^{r*} + \Delta i_{qs}^{r*} \quad (9.169)$$

from which the magnitude of the reference stator current is obtained as follows:

$$i_s^* = \sqrt{i_T^{*2} + i_f^{*2}} \quad (9.170)$$

The flux-producing component of the stator-current command, i_f^*, is zero for $\delta = 90°$, but it is nonzero for δ other than 90°. i_f^* is nonzero when flux-weakening is resorted to, as incorporated later in this section.

The phase angle of the reference stator current is given by

$$\theta_s^* = \delta^* + \theta_r \quad (9.171)$$

where the torque angle is calculated as

$$\delta^* = \operatorname{atan}\left(\frac{i_T^*}{i_f^*}\right) \quad (9.172)$$

The reference stator-phase currents can be generated from i_s^* and θ_s^* by using the transformation equations (9.45):

$$i_{as}^* = i_s^* \sin(\theta_s^*) \quad (9.173)$$

$$i_{bs}^* = i_s^* \sin\left(\theta_s^* - \frac{2\pi}{3}\right) \quad (9.174)$$

$$i_{cs}^* = i_s^* \sin\left(\theta_s^* + \frac{2\pi}{3}\right) \quad (9.175)$$

The reference currents are fed to the inverter, in this particular case illustrated with a hysteresis controller, which makes the actual motor currents follow the commanded values at all times. Current feedback is required for the hysteresis controller to achieve this. A description of the hysteresis current controller can be found in Chapter 4. The reference phase currents are compared to the actual line currents by the hysteresis controller, which in turn controls the switching of the inverter and hence controls the average phase voltages supplied to the PMSM. To operate above the rated speed, flux-weakening is applied. The block labeled as q and d axis currents in Figure 9.43 is elaborated in Figure 9.44 by using a simple flux-weakening control

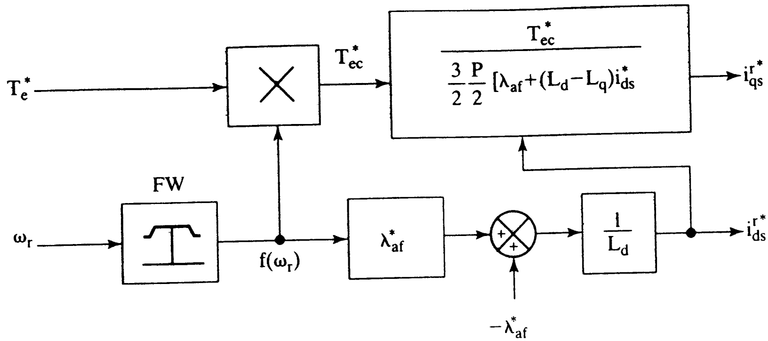


Figure 9.44 Schematic of the q and d axis command current generator of Figure 9.43

strategy. In Figure 9.44, the block labeled as FW (for flux-weakening unit) has the following output:

$$f(\omega_r) = \frac{\omega_b}{\omega_r}; \qquad \omega_b < \omega_r < \omega_{max} \tag{9.176}$$
$$= 1; \qquad 0 < \omega_r < \omega_b$$

If the speed is less than the rated or base speed, the output of block FW is 1, and hence $i_{ds}^* = i_f^* = 0$. If the speed is greater than the rated speed, the flux-weakening component of the stator current is

$$i_{ds}^* = \frac{(f(\omega_r) - 1)\lambda_{af}^*}{L_d} \tag{9.177}$$

Note that $i_{ds}^* < 0$ for flux-weakening. The q axis reference current, i_{qs}^r*, is obtained as follows:

$$T_{ec}^* = T_e^*.f(\omega_r) \tag{9.178}$$

$$i_{qs}^r* = \frac{T_{ec}^*}{\frac{3}{2}\frac{P}{2}[\lambda_{af} + (L_d - L_q)i_{ds}^*]} \tag{9.179}$$

This completes the algorithm for the q and d axis command current generator blocks shown in Figure 9.43.

9.9.3.2 Performance. Dynamic simulation results of the drive system with parameter compensation shown in Figure 9.43 are presented in this section.

Torque-Drive Performance: Figure 9.45 shows the simulations for a step change of rotor flux linkages in the uncompensated and compensated torque-drive system. The system starts with nominal rotor flux linkages; after t = 0.02 s, λ_{af} is changed to 85% of its nominal value and the corresponding effects are studied. It is observed that, in the uncompensated system, since $\Delta i_{qs}^r* = 0$, i_s* does not change, thereby

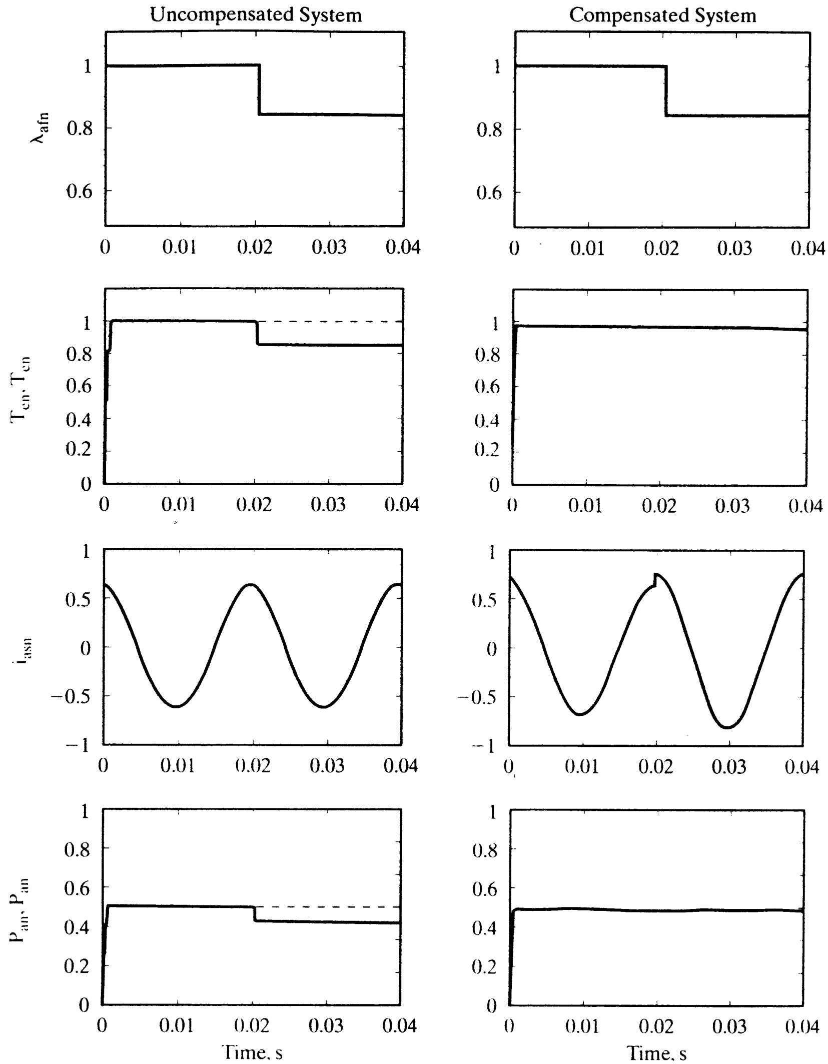


Figure 9.45 Simulation results of the constant-torque-angle-controlled torque drive system with and without parameter compensation when a step change in the rotor flux linkages is applied

reflecting a lack of changes in phase currents due to this step change in rotor flux linkages. Therefore, a decrease in the electromagnetic torque from its reference value is seen, resulting in a corresponding decrease in the actual air gap power. In the compensated system, it is noticed that torque rises to match its reference after initially dropping to a lower value. A corresponding rise in the value of i_{as}, the phase a current, is seen. In this torque-drive system, the rotor speed is assumed to be constant at $\omega_r = 0.5$ p.u. which is the rated speed for the machine under consideration.